

Uncertainty-aware co-phasing of variants and methylation with LongPhase 2

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Abstract

Read-based phasing resolves haplotypes from long reads, yet current methods phase only single-nucleotide variants, weigh all read evidence equally and report no measure of confidence, so that switch errors propagate unnoticed into downstream analyses. Here we present LongPhase 2, which co-phases SNVs, indels, structural variants and allele-specific methylation in a single weighted graph, calibrates every observation by base quality and sequence context, and attaches a phasing entropy to each variant. An optional graph neural network inspects high-entropy sites and removes unreliable phase assignments without altering genotypes. On HG002 nanopore data, LongPhase 2 reduces switch errors 3–5-fold relative to WhatsHap and HapCUT2 while running 6–8 times faster, extends haplotype blocks by up to 70% through co-phasing, and reveals that GRCh38-based benchmarks conceal most of the errors exposed by telomere-to-telomere truth sets. LongPhase 2 is a drop-in replacement for the version already embedded in Clair3, ClairS, Severus and EPI2ME workflows.

Introduction

Haplotype phasing, the assignment of heterozygous variants to their parental chromosomes, is essential for interpreting the diploid human genome^{1,2}. Whether two variants act in *cis* or in *trans* determines compound heterozygosity in recessive disease; allele-specific expression and methylation report imprinting and *cis*-regulatory variation; and the haplotype context of somatic mutations reveals clonal structure and loss of heterozygosity in tumours^{3,4}. Long-read sequencing has made read-based phasing practical at clinical and population scale^{5–9}: individual reads span tens to hundreds of kilobases, connect distant heterozygous sites directly and carry native base-modification signals, yielding haplotype blocks orders of magnitude longer than short reads allow without trios, chromosome-conformation or single-cell strand data^{10,11}.

Read-based phasing is classically formulated as minimum error correction and solved by dynamic programming or graph partitioning, as in WhatsHap^{12,13}, HapCUT2¹⁴ and Margin¹⁵, all of which were designed for single-nucleotide variants (SNVs). We previously introduced LongPhase¹⁶, which recasts phasing as a weighted allele graph, resolves haplotypes at chromosome scale and co-phases SNVs with structural variants (SVs)^{17–19}. It has since become infrastructure rather than an alternative: it is the phasing and haplotagging engine of the Clair3 variant-caller family^{20–22}, of the somatic callers ClairS and ClairS-TO^{23,24} and of DeepVariant local haplotagging²⁵; it supplies the haplotagged alignments consumed by the somatic SV caller Severus²⁶; and

it is embedded in the Oxford Nanopore EPI2ME and PacBio HiFi production workflows^{27,28}. Downstream studies inherit its phase across cancer evolution, rapid rare-disease diagnostics, imprinting, regulatory fine-mapping and immune-gene typing^{29–44}, and it is the baseline against which new phasing algorithms are evaluated^{45–48}. A systematic phasing error therefore does not stop at the phaser; it propagates silently into every analysis built on it.

Deployment at this scale has exposed three limitations shared by all current phasers. They are SNV-centric. Small indels, SVs and 5-methylcytosine (5mC) each carry haplotype linkage that SNVs cannot supply, particularly where heterozygosity is low and blocks terminate, yet methylation-aware methods such as NanoMethPhase⁴⁹, MethPhaser⁵⁰ and HapBridge⁵¹ operate as post-processing on an SNV phaser’s output, and joint small- and structural-variant phasing has been restricted to high-accuracy HiFi reads⁴⁵; no method places all four evidence classes in one model. They weight read evidence uniformly, although the errors that break haplotypes are not uniform: base-calling errors concentrate in homopolymers and tandem repeats, misplaced reads in segmental duplications create spurious linkages, and heterozygous calls in copy-number-altered or loss-of-heterozygosity regions systematically mislead the graph. The result is the switch error, a flip of haplotype assignment within a block that inverts every inference downstream of it, from compound-heterozygosity calls to haplotype-aware somatic detection^{23,26}. And they report phase as a binary state, phased or unphased, with no per-variant confidence, so a phase resting on many concordant reads cannot be distinguished from one resting on two conflicting ones. Learning-based phasers^{46,47} replace the haplotype-assembly algorithm rather than diagnose its errors, and remain single-class and unquantified. These limitations have been underestimated because phasers are benchmarked against GRCh38-based Genome in a Bottle truth sets^{52,53} that exclude much of the repeat-rich sequence in which uniform evidence weighting fails, whereas telomere-to-telomere benchmarks for HG002^{54–56} now assess it.

Here we present LongPhase 2, which addresses these limitations through two coupled advances (Fig. 1). First, LongPhase 2 co-phases SNVs, small indels, SVs and 5mC modifications within a single weighted phasing graph in one pass over the alignments, calling allele-specific methylation directly from modification tags and calibrating every read observation by base quality and by sequence-context reliability, including homopolymer, tandem-repeat and copy-number-aware down-weighting. Second, it attaches to every phased variant a phasing entropy computed from the weighted haplotype votes, and uses that entropy to select sites for an optional graph neural network that combines local attention over the phasing graph^{57,58} with window-wide attention^{59,60}. Variants predicted to be misphased are unphased rather than flipped and the affected phase set is split, so that genotypes are preserved and the corrected output can contain no phase the graph did not support. To our knowledge, a per-variant measure of phase confidence, here an entropy together with a learned error probability and analogous to the genotype quality that variant callers have long reported, has not previously been available from a read-based phaser. LongPhase 2 is a germline phaser and assumes a diploid genome with balanced allele fractions. Tumour genomes, in which loss of heterozygosity, aneuploidy, subclonality and normal-cell admixture violate that assumption, require a different model and are treated separately: we have extended the LongPhase 1 graph to paired tumour–normal somatic haplotyping and tumour DNA fraction estimation in LongPhase-S, and to the tumour-only setting

in LongPhase-TO (companion manuscripts in preparation)^{61,62}. Neither addresses multimodal germline co-phasing or phase confidence, which are the subject of this work.

On HG002 nanopore R10.4.1 data from 10× to 60×, LongPhase 2 reduces switch errors relative to WhatsHap and HapCUT2 by several fold[verify: ~3–5× under GIAB v5; ~1.1–1.3× under v4.2.1] and lowers Hamming distance under both the GRCh38-based GIAB v4.2.1 and the telomere-to-telomere-based benchmarks, while remaining 5–10 times faster[verify from runtime panel]; the divergence between the two truth sets is itself a result, since the GRCh38-based benchmark conceals most of the errors that separate the methods. Co-phasing indels, SVs and methylation extends haplotype blocks beyond the SNV-only limit[verify: N50 ~5 Mb vs ~3 Mb at 60×], and the uncertainty module further reduces switch errors in every variant-class combination, with 86–96% of the calls it unphases being sites that the benchmark does not support as heterozygous[verify]. LongPhase 2 is implemented in C++ with the trained network embedded in the binary, requires no additional dependencies, and is a drop-in replacement for the LongPhase versions already deployed in Clair3, ClairS, Severus and the EPI2ME workflows, so that longer, more accurate and now self-assessed haplotypes reach those pipelines without any change to their interfaces.

Results

LongPhase 2 places SNVs, indels, SVs and methylation in one phasing graph and scores its own confidence

LongPhase 2 resolves haplotypes in three stages (Fig. 1). All heterozygous evidence carried by the same long reads, comprising SNVs, small indels, SVs and allele-specific 5mC sites called by the built-in modcall module, enters a single weighted phasing graph per chromosome, in which each observation is calibrated by base quality and by the reliability of its sequence context so that homopolymers, tandem repeats and copy-number-altered regions contribute less (Fig. 1a,b). For every phased variant the graph reports the weighted votes for the two haplotypes and their normalized Shannon entropy, the phasing entropy (PE), which ranges from 0 for unanimous support to 1 for equivocal support. An optional graph neural network then inspects a local window around every high-entropy variant (Fig. 1c); variants it predicts to be misphased are unphased rather than flipped, and any phase set that becomes disconnected is split, so that genotypes are preserved and no new phase assignment is ever introduced. The outputs are phased VCFs for every variant class and haplotagged alignments (Fig. 1d).

We evaluated LongPhase 2 on HG002 nanopore R10.4.1 whole-genome data down-sampled from 10× to 60× coverage (Methods). Throughout, values quoted at 10–20× are means over ten independent down-sampling replicates, and values quoted at 30–60× come from a single replicate[report s.d. or range for every 10–20× value quoted below, and add error bars/bands to Figs. 2–5; state explicitly in the text or a supplementary table how much of the 30–60× ranking is reproducible, e.g. by adding two further seeds at 30× and 60×]. Phasing accuracy was scored against two independent truth sets: the GRCh38-based Genome in a Bottle (GIAB) v4.2.1 benchmark and the telomere-to-telomere (T2T)-based HG002 Q100 benchmark (referred to here as v5.0q), which extends evaluation into difficult regions that v4.2.1 excludes. Unless

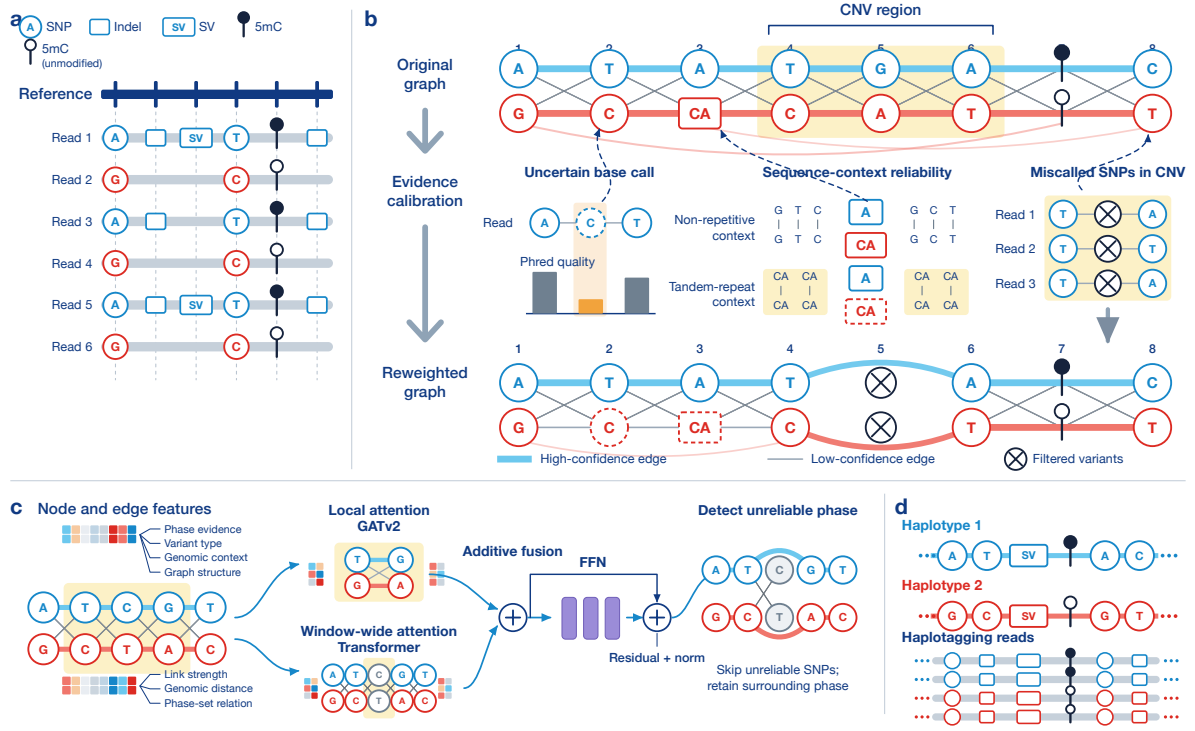


Figure 1: **Overview of LongPhase 2.** **a**, Heterogeneous molecular evidence (SNVs, small indels, structural variants and 5mC calls) from the same long reads is integrated into a single haplotype-resolution problem. **b**, Weighted phasing graph. Allele nodes are connected by read-supported edges; evidence is calibrated by base quality and sequence-context reliability (homopolymers, tandem repeats, miscalled heterozygous sites in copy-number-altered regions) before multi-neighbour haplotype voting and read-based correction. **c**, Optional uncertainty correction. Per-variant phasing entropy selects loci for a graph neural network that combines local attention over the phasing graph (GATv2) with window-wide attention (Transformer); variants predicted to be misphased are unphased and the affected phase set is split, preserving the genotype. **d**, Outputs: phased VCFs for all variant classes and haplotagged alignments.

stated otherwise, results refer to the v5.0q benchmark and to SNVs, the only variant class that all compared tools phase.

LongPhase 2 phases SNVs 3–5-fold more accurately and 6–8-fold faster than WhatsHap and HapCUT2

On identical Clair3 variant calls and SNV-only phasing, LongPhase 2 made several-fold fewer switch errors than WhatsHap 2.8 and HapCUT2 1.3.4 at every coverage (Fig. 2). The switch error rate of LongPhase 2 without the neural network was approximately 0.045% at 30× and 0.035% at 60×, compared with 0.115% and 0.085% for WhatsHap and 0.15% and 0.11% for HapCUT2, a 2.5-fold reduction relative to WhatsHap and a 3.2-fold reduction relative to HapCUT2 (Fig. 2b)[exact rates from TSV; the two fold-changes are coverage averages (WhatsHap 2.6× at 30× and 2.4× at 60×; HapCUT2 3.3× and 3.1×) — decide whether to quote ranges instead, as elsewhere in this section]. Applying the neural network reduced the rate further, to approximately 0.032% at 30× and 0.022% at 60×, so that LongPhase 2 with correction made 3.6–3.9-fold fewer switch errors than WhatsHap and 4.7–5.0-fold fewer than HapCUT2 across 30–60×, and 2.6-fold and 3.1-fold fewer at 10×[quote the WhatsHap and HapCUT2 switch error rates at 10× as well, so that the 2.6- and 3.1-fold claims are verifiable from the reported numbers]. Correction helped most where evidence is sparsest in absolute terms: at 10× the rate fell by 0.038 percentage points, from 0.125% to 0.087%, against 0.013 percentage points at both 30× and 60×[in relative terms the reduction is in fact smallest at 30× (29%) and largest at 60× (37%); confirm which framing the exact numbers support]. The block-wise Hamming distance, which counts misassigned variants rather than switch points and therefore penalizes long-range flips, showed the same ordering: at 30× it was approximately 2.6% for WhatsHap and HapCUT2, 2.4% for LongPhase 2 and 1.8% after correction, and at 10× it fell from 7.7–7.8% for the two competing tools to 5.3% after correction (Fig. 2c).

The accuracy gain cost neither completeness nor haplotype length. All tools phased a similar fraction of benchmark heterozygous SNVs, rising from roughly 78% at 10× to a plateau above 91% from 20× onwards, with WhatsHap phasing about one percentage point more than LongPhase 2 at every coverage (approximately 92.5% versus 91.6% at 30–60×), consistent with the more conservative read-based correction of LongPhase 2 (Fig. 2a). Block N50 grew almost linearly with coverage for all tools and was highest for LongPhase 2 (approximately 2.9 Mb at 60×, versus 2.75 Mb for WhatsHap and HapCUT2); the neural network shortened blocks only marginally (2.87 Mb at 60×), and only at 10–14× did unphasing reduce N50 noticeably (0.84 Mb versus 0.93 Mb at 10×) (Fig. 2d). LongPhase 2 was also substantially faster: phasing took 3 min at 10× and 15 min at 60× on the whole genome, compared with 26 and 95 min for HapCUT2 and 26 and 115 min for WhatsHap, a 6–8-fold speed-up at 30–60×; the neural network added 1–2 min (Fig. 2e)[threads and hardware; exact times; the 6–8-fold figure is derived from the 60× point only—give the 30× times, or restate the range as “at 60×”. Add peak memory alongside runtime (Supplementary table), which Methods already states was recorded].

A telomere-to-telomere benchmark exposes switch errors hidden by the GRCh38 truth set

Scoring the same phased VCFs against the two truth sets changed the apparent separation between methods by more than fourfold (Fig. 3). Under GIAB v4.2.1, all methods converged to a narrow band of 1,600–2,200 switch errors genome-wide from 30× upwards, with LongPhase 2 with correction lowest (approximately 1,650), followed by LongPhase 2 without correction (1,700), WhatsHap (1,800), LongPhase 1.0 (2,000) and HapCUT2 (2,100) (Fig. 3b)[exact counts]. Under v5.0q the same VCFs separated four to five times more widely: at 60× HapCUT2 made approximately 2,330 switch errors, WhatsHap 1,790, LongPhase 1.0 1,380, LongPhase 2 740 and LongPhase 2 with correction 490, and at 10× the corresponding counts were 5,030, 4,390, 4,030, 2,260 and 1,530 (Fig. 3c). The additional errors that WhatsHap and HapCUT2 accumulate therefore lie predominantly in regions that v4.2.1 does not assess, including segmental duplications, satellite arrays and other repeat-rich sequence that the T2T-based benchmark newly covers, and the evidence-calibration steps of LongPhase 2 are most effective precisely there[support this attribution directly: stratify the v5.0q switch errors by genomic context (segmental duplication, satellite, tandem repeat, unique), and add an ablation in which the homopolymer, overlapping-alignment and copy-number-aware filters are disabled one at a time. As written, the causal claim rests on inference rather than on an experiment, and a reviewer will ask for it].

The two benchmarks are far less discriminating on the metrics that do not depend on them. The fraction of benchmark heterozygous SNVs phased (Fig. 3a) and the phase-block N50 (Fig. 3d), which is benchmark independent, separate the five methods much less than the error counts do[quote the phased fraction and N50 of LongPhase 1.0 at 10× and 60×; LongPhase 1.0 appears only in Fig. 3 and is currently never described quantitatively for these two panels]. Hamming distance was also nearly identical under the two benchmarks for HapCUT2, WhatsHap and LongPhase 2 with and without correction (Fig. 3e,f), indicating that the additional v5.0q switch errors are mostly isolated switches rather than long-range flips. LongPhase 1.0 was the exception: it showed a Hamming distance of 6–12% under both benchmarks despite an intermediate number of switch errors, revealing long-range flips within blocks that the read-based correction and copy-number-aware filtering introduced in LongPhase 2 eliminate.

Co-phasing indels, SVs and methylation lengthens blocks by up to 70% but raises the SNV switch error rate

Joint phasing of all four variant classes extended haplotypes well beyond the SNV-only limit at a measurable cost in SNV accuracy, which correction then more than repaid (Figs. 4 and 5). Indels and methylation sites provide new links across SNV-poor regions, and block N50 at 60× increased from approximately 2.9 Mb for SNV-only phasing to 4.3 Mb with indels and 5.0 Mb with all four classes, and to 3.1 Mb with methylation alone (Fig. 5, right column). At the same time, indel observations are noisier than SNV observations: the SNV switch error rate of uncorrected co-phasing rose from 0.036% (SNV only) to 0.048% (all classes) at 60× and from 0.125% to 0.156% at 10×, while the Hamming distance rose from 1.5% to 7.2% at 60× (Fig. 4e,f)[0.036% here versus 0.035% quoted for the identical quantity (uncorrected SNV-only

switch error rate at 60×) in the Fig. 2b paragraph; reconcile against the TSV and use one value throughout]. Decomposing the combinations showed that the additional error is carried almost entirely by indels: SNV+SV and SNV+methylation had switch error rates and Hamming distances indistinguishable from SNV-only phasing, whereas SNV+indel matched the four-class configuration (Fig. 5).

Co-phasing did not reduce how completely each class was phased. The fraction of phased SNVs was unchanged relative to SNV-only phasing (Fig. 4a). The fraction of benchmark heterozygous indels that were phased rose from approximately 30% at 10× to 45% at 60× and was unaffected by the neural network (Fig. 4b)[clarify denominator: all GIAB indels or those called by Clair3]. Of the allele-specific 5mC sites identified by modcall, more than 99% were phased at every coverage, and the neural network unphased 1.5–5% of them, mostly at low coverage (Fig. 4c). Roughly 68–70% of heterozygous SVs called by Sniffles2 were phased, decreasing slightly with coverage as more low-support SVs were called, and correction removed a further 2 percentage points (Fig. 4d).

[Two comparisons are missing from this section and will be requested. (i) MethPhaser, which Methods states was run in the methylation-assisted setting, is never reported: add the SNV+5mC comparison of LongPhase 2 against WhatsHap followed by MethPhaser (block N50, switch error rate, Hamming distance) or remove MethPhaser from Methods. (ii) LongPhase 1.0 already co-phases SNVs with SVs, so the SNV+SV and four-class configurations should be compared against it to separate the contribution of the new evidence classes from that of the new evidence calibration; LongPhase 1.0 currently appears only in Fig. 3. Also report the phasing accuracy, not only the phased fraction, of the indels, SVs and 5mC sites themselves.]

Neural network correction reduces switch errors in every variant-class configuration

The neural network reduced the SNV switch error rate in all five configurations and at every coverage (Fig. 5, left column). For SNV-only phasing the rate fell by 35–40% (for example from 0.036% to 0.022% at 60×); for SNV+indel and for all four classes it fell from 0.048% to 0.027% at 60× and from 0.156% to 0.092% at 10×. Corrected co-phasing was therefore more accurate than uncorrected SNV-only phasing at every coverage while retaining substantially longer blocks (Fig. 4e). The correction was even more effective on the Hamming distance of indel-containing configurations, which fell from 7.2% to 3.3% at 60× and from 12.5% to 7.5% at 12× (Fig. 5, middle column)[confirm the 12× point: every other low-coverage value in this section is quoted at 10×], indicating that a large share of the indel-induced errors are long-range flips that the network recognizes from the surrounding vote structure. The price was a moderate reduction in block N50 for indel-containing configurations, from 4.3 Mb to 3.7 Mb (SNV+indel) and from 5.0 Mb to 4.1 Mb (all classes) at 60×, because unphased variants that bridged two blocks split them; for configurations without indels the N50 was essentially unchanged. The trade-off therefore resolves in favour of co-phasing: even after correction, four-class co-phasing yielded blocks approximately 40% longer than SNV-only phasing at 60× together with a lower switch error rate than uncorrected SNV-only phasing[verify 4.1 vs 2.9 Mb].

[Ablations a reviewer will demand for this section: (i) the network versus the simple baseline

it replaces, namely unphasing every variant with PE above a fixed threshold, matched for the number of variants unphased—this is the control for the claim made in the next section that a fixed entropy threshold cannot substitute for the network; (ii) sensitivity of all metrics to the break threshold (0.30) and to the PE trigger threshold (0.80), ideally as a switch-error-versus-N50 trade-off curve; (iii) precision and recall of the network on held-out switch-error labels. Also state explicitly, here or in Methods, which HG002 chromosomes and coverages were held out of training, since training and evaluation use the same individual.]

Unphased variants are predominantly unsupported heterozygous calls

Most of what the network removes was never real phasing information. We classified every SNV that LongPhase 2 phased and the network subsequently unphased (SNV-only phasing, single replicate, seed 1) against the v5.0q benchmark (Fig. 6). The network unphased approximately 60,000 SNVs at every coverage, but 86–96% of them were calls that the benchmark does not support as heterozygous, either because the site is absent from the benchmark (80–94%) or because it is homozygous there; only 4.0% (2,411 variants) at 60× and 14% at 10× were genuine heterozygous sites (Fig. 6a).

Cross-tabulating each unphased variant by read depth and by phasing informativeness, the number of informative reads linking it to its neighbours, showed that genuine heterozygotes were removed overwhelmingly at normal depth with weak phasing evidence (61–81%), whereas benchmark-absent calls shifted towards low depth as coverage increased (12.5% at 10× to 58.6% at 60×), the signature of hard-to-map regions in which reads are lost to alternative loci (Fig. 6b). No variant was unphased on the basis of high depth alone. Among the genuine heterozygotes that were removed, 43% at 10× had a phasing entropy of exactly zero, that is, unanimous but sparse votes, compared with 17% of benchmark-absent calls, and this share fell to 22% at 60× as evidence accumulated (Fig. 6c); entropy alone would therefore not have identified them, whereas the network, which sees the votes of all neighbours in the window, did[Fig. 6c also plots the $PE \geq 0.8$ category, which the text does not report; add the corresponding prevalences for genuine heterozygous and benchmark-absent calls, or drop the category from the panel]. Conversely, clustering within 50 bp and low genotype quality were present in 46–49% of benchmark-absent calls at every coverage but were far less frequent among genuine heterozygotes, marking them as variant-caller artefacts rather than phasing failures.

The loss of true phasing information caused by correction is therefore small, about 2,400 SNVs at 60× out of [N] phased, and most of what is removed are false heterozygous calls whose spurious phase would otherwise propagate into haplotagged reads and downstream haplotype-aware callers.

[Generalization experiments missing from the Results as a whole, each of which a Nature Methods reviewer is likely to require, and none of which may be asserted without data: (i) a second individual, e.g. HG001 or HG005, to show that the network does not encode HG002-specific structure—the model was trained on HG002 and every result above is on HG002; (ii) PacBio HiFi data, since the Discussion claims the graph algorithm supports both platforms and the Introduction contrasts LongPhase 2 with HiFi-restricted joint phasing; (iii) a runtime and peak-memory table across coverages, variant-class configurations and thread counts, covering

LongPhase 2, WhatsHap, HapCUT2 and LongPhase 1.0; (iv) a demonstration that improved phase propagates downstream, e.g. haplotagging accuracy or haplotype-aware variant calling, which the Discussion asserts as the motivation for the work.]

[Item (iv) above may already be done. The group’s algorithm slides contain a ClairS somatic-calling experiment across six tumour cell lines (COLO829, H1437, H2009, HCC1395, HCC1937, HCC1954) at five simulated purities (20–100%), in which the germline phaser supplying the haplotagged alignments is varied between WhatsHap 2.4, LongPhase 1.7.3 and LongPhase 2.0 and everything downstream is held fixed. This is exactly the downstream-impact result the Discussion promises, and it is the strongest available answer to “does a lower switch error rate actually matter?”: on HCC1395 at 100% purity, somatic SNV F1 rises from 0.8250 with LongPhase 1.7.3 to 0.8398 with LongPhase 2.0, with the margin widening at low purity. Decide whether to bring it into this manuscript as a final Results section, noting three caveats: the comparison against WhatsHap is mixed rather than uniformly favourable (WhatsHap is marginally ahead at 100% purity on HCC1395 and behind at 20–40%), so the claim must be stated as “improves on the LongPhase version ClairS currently embeds” rather than as a win over all phasers; the experiment uses ClairS v0.4.0 in `ssrs` mode, which must be stated; and this is germline phasing evaluated by its effect on somatic calling, which must be framed so that it is not read as a somatic-phasing claim belonging to LongPhase-S or LongPhase-TO.]

[The same slides also contain an HG002 germline phasing comparison (LongPhase 1.7.3, LongPhase 2.0, WhatsHap 2.4 at 10–60×) that does NOT match the experiment reported above: it uses DeepVariant v1.9.0 calls rather than Clair3, the GIAB benchmark rather than v5.0q, and different tool versions (WhatsHap 2.4 vs 2.8; LongPhase 1.7.3 vs the “LongPhase 1.0” named in Fig. 3). Its switch error rates are also nearly flat in coverage (0.056% at both 10× and 60×) whereas Fig. 2 reports 0.125% falling to 0.035%. Reconcile the two runs, or state explicitly which configuration the manuscript reports; and fix the LongPhase version label, which appears as 1.0 in the manuscript and 1.7.3 in the slides.]

Discussion

LongPhase 2 joins two ideas that read-based phasing has kept apart: that every class of heterozygous evidence a read carries belongs in one haplotype model, and that a phaser should report, variant by variant, how far its own output can be trusted. The first extends an observation made independently for methylation^{49–51} and for SVs^{16,45}: SNVs alone strand haplotype blocks wherever heterozygosity is low. Calling allele-specific 5mC from modification tags into the same weighted graph as SNVs, indels and SVs raises block N50 above SNV-only phasing by more than 70% at 60× (5.0 versus 2.9 Mb; 40% after correction), with no post-processing step and on nanopore rather than HiFi reads, to which joint small- and structural-variant phasing had been confined⁴⁵. The second replaces the binary phased-or-unphased state that every current tool reports, ours included, with a graded one, giving downstream analyses the handle on phase reliability that genotype quality provides for calls.

The most consequential finding here concerns evaluation rather than algorithms: scored against two truth sets, the same phased VCFs on the same reference yield not only different error magnitudes but a different ordering of methods. GIAB v4.2.1⁵³ excludes most segmental

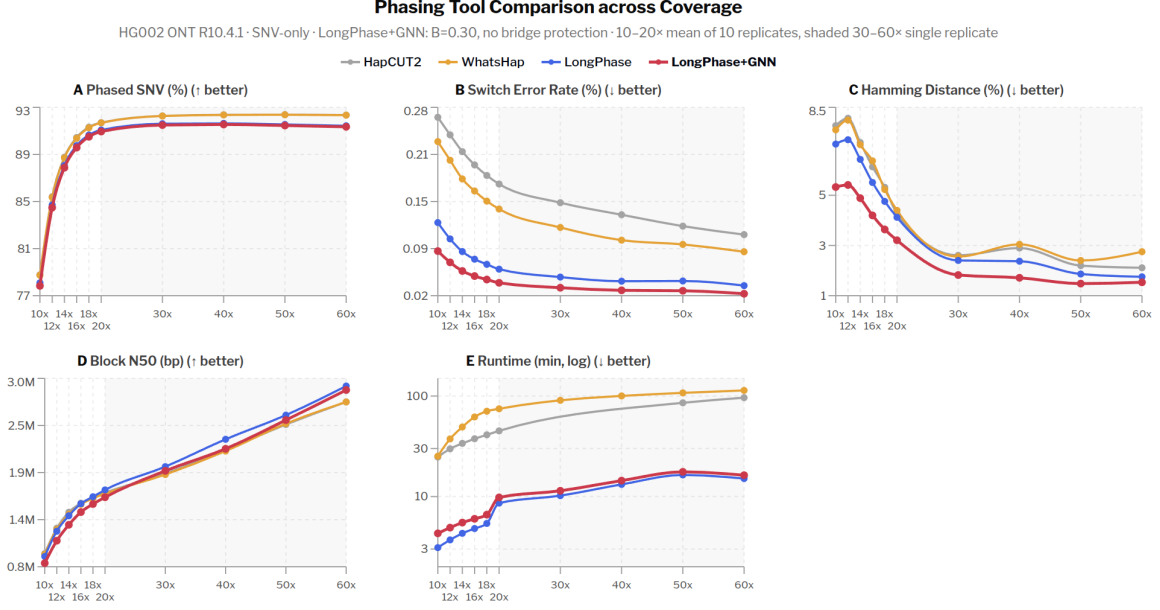


Figure 2: **SNV phasing accuracy, block length and runtime compared with WhatsHap and HapCUT2.** HG002 nanopore R10.4.1 reads down-sampled to 10–60×; SNV-only phasing of identical Clair3 calls; scored against the T2T-based HG002 Q100 (v5.0q) benchmark. Points at 10–20× are means of ten down-sampling replicates; the shaded region (30–60×) is a single replicate. **a**, Fraction of benchmark heterozygous SNVs phased. **b**, Switch error rate. **c**, Block-wise Hamming distance. **d**, Phase-block N50. **e**, Wall-clock runtime (log scale, [t] threads); LongPhase 2 with correction includes the neural-network stage. Grey, HapCUT2 1.3.4; orange, WhatsHap 2.8; blue, LongPhase 2; red, LongPhase 2 with neural-network correction.

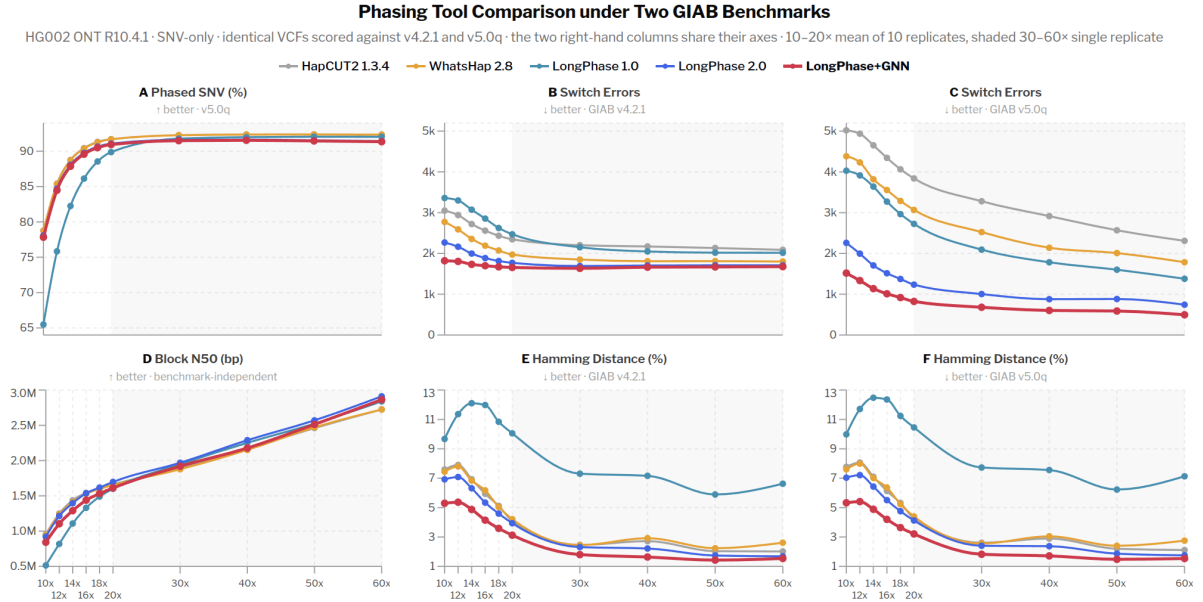


Figure 3: **The same phased VCFs scored against the GRCh38-based GIAB v4.2.1 and the T2T-based v5.0q benchmarks.** SNV-only phasing of HG002 nanopore R10.4.1 data; replicates as in Fig. 2. **a**, Fraction of benchmark heterozygous SNVs phased (v5.0q). **b,c**, Genome-wide switch-error counts under v4.2.1 (**b**) and v5.0q (**c**); the two panels share axes. **d**, Phase-block N50 (benchmark independent). **e,f**, Hamming distance under v4.2.1 (**e**) and v5.0q (**f**). Grey, HapCUT2 1.3.4; orange, WhatsHap 2.8; teal, LongPhase 1.0; blue, LongPhase 2; red, LongPhase 2 with neural-network correction.

duplications, satellite arrays and large tandem repeats from assessment; the T2T-based HG002 Q100 benchmark^{54–56}, projected onto GRCh38 from a near-complete diploid assembly of the same individual, assesses exactly that sequence. Every tool accumulates more switch errors there, in rate as in absolute terms, and the spread between tools widens from under 30% to three- to fivefold: mis-mapped reads, homopolymer errors and collapsed paralogues create systematically wrong linkages there rather than random noise, the errors that the evidence calibration in LongPhase 2 targets. The ranking can also invert: LongPhase 1.0 made more switch errors than WhatsHap under v4.2.1 and fewer under Q100. Almost every published comparison of phasers^{12,14–16,45,50} used GRCh38-based GIAB releases and so describes the easier part of the genome. Two observations argue that the extra errors are genuine and not artefacts of a draft truth set: Hamming distance is nearly unchanged between benchmarks for the four current tools, marking isolated switches rather than the long-range disagreements a mis-projected benchmark would create, and the advantage of LongPhase 2 over WhatsHap and HapCUT2 persists, with a smaller margin, under v4.2.1. Q100 is nonetheless still being refined, and residual discordance in the hardest regions may be benchmark rather than phasing error; comparisons should therefore report both truth sets and stratify switch errors by genomic context^{63,64}.

The neural network complements the graph algorithm rather than replacing it. Where Ralph⁴⁶ and GCphase⁴⁷ learn haplotype assembly itself and must scale to whole chromosomes, LongPhase 2 confines learning to windows around high-entropy variants, as graph networks have been used to resolve ambiguous regions of assembly graphs rather than to assemble genomes⁶⁵; inference costs one to two minutes per genome and fewer than 700,000 parameters compile into the binary. Because the correction only withholds phase, the objection arises that fewer switch errors merely reflect a smaller denominator. Three observations argue otherwise. The rate is computed per assessed pair, and of the roughly 60,000 SNVs unphased at 60× only 2,411 are heterozygous in the benchmark and enter that denominator, far too few to explain the fall from 740 to 490 genome-wide switch errors. Block N50 is almost unaffected (2.87 versus 2.9 Mb), which pruning informative links could not achieve. And the phased fraction stays within about a percentage point of WhatsHap (91.6% versus 92.5% at 30–60×)[confirm 91.6% is the phased fraction after correction; this argument rests on it]. At 10× the trade is less favourable, 14% of what is unphased being genuine heterozygotes and N50 falling from 0.93 to 0.84 Mb. The asymmetry is the point: 86–96% of what is removed are calls the benchmark does not support as heterozygous, mostly caller artefacts at clustered or low-quality sites whose spurious phase would otherwise reach haplotagged reads and the somatic callers that consume them^{23,24,26,66}.

Several limitations remain. The network was trained and evaluated on one individual, HG002, sequenced on a single nanopore chemistry[Methods must state which HG002 chromosomes and coverages were held out of training, and which truth set supplied the training labels; without a chromosome-level hold-out the evaluation is circular]. Two choices bound the overfitting: the features are generic (vote counts, entropy, sequence context, graph topology; no sequence embeddings, no sample-specific statistics), and the output is restricted to withholding phase, so a miscalibrated model costs contiguity rather than inventing haplotypes. Transfer to PacBio HiFi, duplex chemistry and non-human genomes is untested; the graph algorithm supports both platforms, but for the network reported here cross-platform performance is an

expectation, not a result. Indel co-phasing is a trade-off we have chosen rather than solved: it lengthens blocks but raises the uncorrected SNV switch error rate (0.036% to 0.048% at 60×), and is worth taking only because correction more than repays the loss, leaving four-class co-phasing both more accurate and 40% longer-blocked than uncorrected SNV-only phasing. Better indel calling^{21,25} would dissolve it, and SNV-only phasing remains available meanwhile. The phased fraction of SVs and indels is bounded by the upstream callers, and 5mC co-phasing requires modification-aware base calling and MM/ML tags retained through alignment. Finally, LongPhase 2 assumes a diploid genome with balanced allele fractions, which tumours violate. We address that setting separately, in companion manuscripts that carry the LongPhase 1.0 graph, rather than the germline method described here, into tumour genomes: LongPhase-S⁶¹ for paired tumour–normal somatic haplotyping and tumour DNA fraction estimation, and LongPhase-TO⁶² for somatic haplotype reconstruction from a tumour alone (both under review; Methods). The division of labour is deliberate — germline multimodal co-phasing and phase confidence here, somatic lineage and loss of heterozygosity there — and it sets the next step, since tumour analyses built on germline blocks, from somatic callers to copy-number reconstruction⁶⁷, inherit whatever phase error the germline phaser leaves behind.

Trio assembly^{68,69}, Hi-C⁷⁰ and Strand-seq^{10,71} reach chromosome scale but need extra samples or libraries, and imputation^{3,72,73} depends on panels still sparse for structural variation and many populations; LongPhase 2 needs one sample and one library, and phase reliability makes its output a natural input to hybrid read-plus-panel strategies⁶⁴. As a drop-in replacement in pipelines already built on LongPhase^{20,23,26,27}, it delivers the new annotations without interface changes, and what they enable is concrete. A compound-heterozygous diagnosis rests on a single *trans* call between two variants in one gene; the entropy of those variants and of the links between them turns that call into a quantity a diagnostic report can qualify, flagging the cases that warrant a confirmatory assay. Allele-specific methylation analyses treat every haplotagged read alike, whereas weighting reads by the confidence of the variants that tagged them would keep imprinting and *cis*-regulatory signals from being diluted by thinly supported blocks. And haplotype-aware somatic callers^{23,26} accept inherited phase without qualification; propagating entropy would let them discount precisely the loci at which switch errors concentrate. Like genotype quality before it, phase confidence becomes useful only once it is consumed.

Methods

Overview of LongPhase 2

LongPhase 2 takes as input long-read alignments (BAM or CRAM), a reference genome and heterozygous variant calls, and produces phased VCF files for every variant class together with haplotagged alignments (Fig. 1). The pipeline consists of four sub-commands implemented in a single C++ binary linked against HTSlib⁷⁴: **modcall** identifies allele-specific 5-methylcytosine (5mC) sites from the MM/ML modification tags of the alignments; **phase** co-phases SNVs, small indels, SVs and 5mC sites in one weighted phasing graph per chromosome and emits per-variant phasing diagnostics; the optional **gnn** sub-command applies a graph neural network to the diagnostics and removes phase assignments that are predicted to be erroneous; and **haplotag**

assigns each read to a haplotype using the phased variants. A fifth sub-command, `compare`, implements the evaluation metrics (below). `phase`, `modcall` and `compare` are parallelized over chromosomes with OpenMP; `gnn` distributes trigger windows over a pool of C++ threads, sharing one immutable copy of the network weights. The graph phasing algorithm extends the design of LongPhase 1¹⁶; the sections below describe the complete current algorithm so that the manuscript is self-contained.

LongPhase 2 is a germline, diploid phaser; the somatic setting is outside its scope and is treated in two companion manuscripts from this laboratory that extend the LongPhase 1.0 graph to tumour genomes (LongPhase-S⁶¹, preprint under review; LongPhase-TO⁶², manuscript in review). Neither is a dependency of the present work: no code path, trained model, data set or benchmark is shared with them, and none of the results reported here requires either of them. The scope boundary is set out in the Introduction and Discussion.

Input variant calls

Heterozygous SNVs and small indels were taken from a standard long-read small-variant caller (Clair3²⁰ in this study); homozygous and filtered records are carried through to the output unchanged but do not enter the graph. With `-indels`, small indels were phased alongside SNVs; indels with `QUAL` below a user threshold (`-indelQuality`, default 0, i.e. disabled) were excluded and marked in the output `FILTER` column. Each indel was additionally annotated as lying in a tandem repeat if its reference context showed a repeated unit consistent with the indel length; such tandem-repeat indels were retained but down-weighted throughout (see below). SVs were taken from Sniffles2¹⁸ or cuteSV¹⁹ run with read-name reporting (`-output-rnames` or `-report_readid`); homozygous SVs, SVs whose breakpoint coincides with an SNV, and SVs with duplicated positions were excluded. For each retained SV we recorded the set of supporting read names and the SV length.

Calling allele-specific methylation (`modcall`)

Base-level modification probabilities (Supplementary Fig. 9) were read from the MM/ML tags produced by Dorado⁷⁵ (or the PacBio equivalent) and carried into the alignments by `minimap2 -y`⁷⁶. For each CpG on each read, the base was labelled methylated if the modification probability was ≥ 0.8 (`-modThreshold`) and unmethylated if it was ≤ 0.2 (`-unModThreshold`); intermediate values were treated as noise. Reads on the reverse strand were mapped to the forward-strand CpG coordinate and the two strands of each CpG were genotyped jointly. Each read was thereby assigned to one of three states at the site, and a site was called heterozygous (allele-specifically methylated) when the two states were balanced and the ambiguous fraction small:

$$\frac{\min(M_R, U_R)}{\max(M_R, U_R)} \geq 0.6 \text{ (-heterRatio)}, \quad \frac{A_R}{M_R + U_R + A_R} \leq 0.2 \text{ (-noiseRatio)}, \quad (1)$$

where M_R , U_R and A_R are the numbers of methylated, unmethylated and ambiguous reads; otherwise the site was called homozygous methylated or unmethylated and excluded from phasing. Only CpG dinucleotides were considered; modification calls at non-CpG cytosines were

discarded.

A site passing equation (1) is balanced but not necessarily allele-specific, because a site at which methylation varies stochastically between reads satisfies the same inequalities. Candidate sites were therefore required to co-segregate with a neighbouring heterozygous marker. Writing RR , RA , AR and AA for the numbers of reads carrying each of the four combinations of the two states at a pair of sites, a pair was accepted as linked when

$$\frac{\max(RR + AA, RA + AR)}{RR + AA + RA + AR} \geq 0.9 \text{ (-connectConfidence)}, \quad (2)$$

that is, when nearly every read supports one of the two possible pairings rather than a mixture. Equation (2) is the same quantity that the phasing graph later accumulates as P and Q , evaluated here on read counts instead of quality-weighted edges. Linking was applied in two modes. In *SNV-anchored* mode, used whenever an SNV VCF was supplied, a CpG was accepted as a strong anchor if its methylation state satisfied equation (2) against at least one SNV within the next 20 variants (`-connectAdjacent`) on at least $\max(6, (d_1 + d_2)/4)$ reads, where d_1 and d_2 are the read depths of the two sites. In *methylation-only* mode, used where no informative SNV is within range — which is precisely where methylation is most valuable for bridging — the same criterion was applied between two candidate CpGs. Weakly connected CpGs were accepted only if they satisfied equation (2) on at least six reads against an already accepted CpG, in an iterative expansion of two rounds (`-iterCount`, default 2). Consecutive CpG positions (adjacent coordinates) were merged and represented by their first position. The resulting VCF records the methylated and unmethylated read names (MR, NR), the read strand and the per-site depths, and is passed to `phase` through `-mod-file`.

The values quoted here are the compiled-in defaults (`ModCall.cpp`, lines 67–73 at commit cc17fb1). [The `modcall -help` text at the same commit advertises 0.7 for `-heterRatio` (`ModCall.cpp:29`) and 6 for `-connectAdjacent` (`ModCall.cpp:32`), which contradict the code defaults of 0.6 and 20 used above and in Supplementary Table 5. Fix the help text, then confirm which values were actually used for the benchmarks.] [Two further `modcall` CLI defects at cc17fb1 that will confuse anyone reproducing this work: the long option registered for the BAM input is `-methylbamfile`, not the `-bam-file` shown in the help text (`ModCall.cpp:49`), and `a`, `c` and `k` are absent from the `shortopts` string (`ModCall.cpp:38`), so `-a`, `-c` and `-k` do not parse although the long forms do. Either fix these or document the working spellings.]

Extraction of read-level allele observations

For each chromosome, all alignments with mapping quality ≥ 1 (`-mappingQuality`) that were not unmapped, secondary or marked as optical/PCR duplicates were streamed once and, for every variant they overlap, converted into an allele observation with an associated quality. Supplementary alignments were deliberately retained as phasing evidence (`ParsingBam.cpp`, `BamParser::isValidAlignment`, lines 945–961), which is why the overlapping-alignment filter below is required; this differs from the stricter filters applied inside `modcall` and `haplotag`. SNV alleles were read from the aligned base and inherit its Phred base quality. Insertion and deletion alleles were called from the CIGAR string: a read supports the ALT allele if a CIGAR insertion

or deletion begins immediately after the variant position, and the REF allele otherwise; when a read deletion spans an SNV in a homopolymer of length ≥ 3 , the SNV allele was recovered from the base following the deletion. SV alleles were assigned by scanning a window of 20 CIGAR operations (`-svWindow`) around the SV start for an insertion or deletion whose length is within 10% of the SV length (`-svThreshold`); reads listed by the SV caller as supporting the SV were also assigned the ALT allele. 5mC alleles were taken from the `modcall` read lists (methylated = allele 0, unmethylated = allele 1) restricted to reads whose strand matched the recorded strand. Soft- and hard-clipping operations longer than 5 bp were counted per reference position, separately for clips at the front (left) and back (right) end of an alignment, as evidence of alignment breakpoints for copy-number detection (below).

[Flag-name discrepancy, `Phasing.cpp` at ccl7fb1: the usage text advertises `-w`, `-sv-window` and `-h`, `-sv-threshold` (lines 40–41) but the `longopts` table registers `-svWindow` and `-svThreshold` (lines 78–79), so the hyphenated long forms do not parse. The same mismatch exists in `Haplotag.cpp` (usage lines 23–24 vs `longopts` lines 52–53). This Methods text and Supplementary Table 4 now use the spellings that actually work; either fix the `longopts` tables or fix the help text, and then make the manuscript match the release.]

Three filters were applied before graph construction. (1) *Homopolymer SNV filter*: when two SNVs both lie in homopolymers of length ≥ 3 and are ≤ 2 bp apart, the downstream SNV was removed, because such pairs are dominated by systematic homopolymer-length errors. [Earlier drafts described this filter as nanopore-specific. It is not: `SnpParser::filterSNP` (`ParsingBam.cpp`:584–648) contains no `isONT` test, and `PhasingProcess.cpp`:123 calls it unconditionally whenever a reference is supplied, so it also applies under `-pb`. Decide whether the platform gate was intended; if it was, restore it in the code, and if it was not, say so here and in Supplementary Fig. 1b, whose legend also calls it nanopore-specific.] (2) *Overlapping-alignment filter*: when two alignments of the same read overlapped by more than 20% of their combined span (`-overlapThreshold`), the shorter alignment was discarded. (3) *Copy-number-aware filter* (Supplementary Fig. 4). Collapsed paralogues and copy-number-altered segments recruit reads from more than one locus, so that paralogue-specific differences are called as heterozygous variants and link alleles that are not on the same chromosome. Such segments are bounded by alignment breakpoints, and the two boundary shapes they produce were detected from the per-position front- and back-clip counts by a finite-state scanner with two entry conditions. A *step* boundary, characteristic of a copy-number transition, was declared when ≥ 5 front-clips occurred at a single position; a *ramp* boundary, characteristic of the progressively nested breakpoints left by breakage–fusion–bridge cycles, was declared when front-clips merely exceeded back-clips at a position without reaching that threshold. In both states the scanner carries a running excess c of front- over back-clips and extends the candidate end by a 30 kb look-ahead while $c > 30$. A step interval was closed when the back-clip count at a position reached an adaptive pull-down threshold — $n/2$ where n is the front-clip count that opened the interval, when $n \geq 10$, and 5 otherwise — or when c decayed to an adaptive floor (5 for $n \geq 20$, $n/4$ for $10 \leq n < 20$, and 2 for $n < 10$). A ramp interval was closed when the back-clip count reached $c/4$ (or 5 while $c \leq 20$), so that the gradual accumulation is terminated by the drop that ends it. Candidates exceeding 200 kb, or in which c fell to zero, were abandoned.

Within each interval, every read was scored by its *mismatch load* R_m , the number of heterozygous sites inside the interval at which the read carries the alternate allele, and every heterozygous variant v in the interval was then scored by

$$\text{MR}(v) = \frac{\bar{R}_m^{\text{ALT}}}{\bar{R}_m^{\text{REF}} + \bar{R}_m^{\text{ALT}}}, \quad (3)$$

where $\bar{R}_m^{\text{ALT}}$ and $\bar{R}_m^{\text{REF}}$ are the mean loads over the reads supporting the alternate and the reference allele of v . Variants with $\text{MR}(v) \geq 0.7$ were removed from all reads. The statistic separates the two cases that matter: at a genuine heterozygous site both allele populations are ordinary reads of the locus and carry comparable loads, so $\text{MR} \approx 0.5$, whereas at a paralogue-specific difference the alternate allele is carried by reads that originate elsewhere and therefore mismatch the reference at many other sites in the same interval as well, driving MR towards 1. A variant was scored only when both means were non-zero, so that sites at which one allele population carries no other alternate allele at all are never removed on this basis. Equation (3) and the 0.7 threshold are implemented in `VairiantGraph::calculateCnvMismatchRate` (`PhasingGraph.cpp:519–557`, test `variant.allele == 1` at line 550), `calculateAverageMismatchRate` (lines 605–642) and `filterHighMismatchVariants` (lines 645–690); the two-state scanner is `Clip::getCNVInterval` (lines 1128–1229) with the adaptive thresholds set in `Clip::updateThreshold` (lines 1112–1127). [An earlier draft called the per-read score “mismatches to the phasing graph”, which was wrong; it is the count of alternate alleles the read carries inside the interval, i.e. its mismatch load against the reference, and the text and Supplementary Fig. 4 now say so. Confirm that this is the intended statistic.] [The 0.7 threshold in equation (3) is presented here without evidence, but the evidence exists: the TP/FP histogram of $\text{MR}(v)$ on HG002 (true heterozygous sites concentrated below 0.7, false calls piled at 0.8–1.0) is already in the group’s algorithm slides and should be added as Supplementary Fig. 4c with the threshold marked. Regenerate it on the Clair3 calls and coverages used in this paper, since the slide version used DeepVariant v1.9.0 calls. Also report how many intervals are detected genome-wide, their total span, and how many variants the filter removes at each coverage.]

Weighted phasing graph

Each heterozygous variant v contributes two nodes (Supplementary Figs. 1–3), v^{ref} and v^{alt} . For every read, the observed alleles were sorted by position and each allele node was connected to the allele nodes of the next $k = 35$ variants on the same read (`-connectAdjacent`), provided the two variants were within 300 kb (`-distance`). Each read contributes a weight of 1 to the edge if both of the two observations have Phred quality ≥ 12 (`-baseQuality`) and a weight of 0.1 (`-edgeWeight`) otherwise. SV observations from the caller’s read list were given a fixed high quality, whereas reads not listed by the caller were treated as REF with a lower quality; 5mC and indel observations were given a fixed high quality. For a pair of variants (u, v) this yields four accumulated weights $w_{rr}, w_{ra}, w_{ar}, w_{aa}$, the total support for the two allele pairings $P = w_{rr} + w_{aa}$ (*cis* between reference alleles) and $Q = w_{ra} + w_{ar}$, and their similarity $s = \min(P, Q) / \max(P, Q)$.

Haplotype resolution by multi-neighbour voting

Variants were processed in coordinate order. For the current variant u , each of its k downstream neighbours v received a vote for the pairing with the larger support (P versus Q). A pair was declared uninformative and cast no vote if $s > 0.7$ (`-edgeThreshold`); for SNV-5mC pairs the threshold was 0.3, and such pairs cast no vote at all when their total accumulated support was below 1. Votes were weighted: 1 by default; 20 when the pairing was near-unanimous ($s \leq 0.1$ with total support ≥ 1 , or one pairing accumulated less than 1 unit of support while the other accumulated at least 1); and 0.1 when the voting variant was a tandem-repeat indel. The vote was translated into a haplotype vote for v using the already assigned haplotype of u , and the two weighted haplotype totals $h_1(v)$ and $h_2(v)$ were accumulated from all upstream voters. A guard against a single erroneous ultra-long read counted, among the votes received by v , those whose total support was at most 1 read-equivalent; when more than three such votes existed and at least one qualifying vote remained, h_1 and h_2 were recomputed from only those votes with $s < 0.2$, weight ≥ 1 and a voting variant that is not a small indel. Variant v was assigned to the haplotype with the larger total. When $h_1 = h_2$ (including the case of no votes at all) and v lay beyond the last position reached by any accepted edge, a new phase block was opened at v ; a tie at a position already covered by an accepted edge left the previous block intact. Blocks were identified by the position of their first variant, which defines the PS tag.

Read-based correction

After the initial assignment (Supplementary Fig. 5), every read was assigned to a haplotype by weighted majority over its phased alleles, with SNV and SV alleles contributing 1, indel and tandem-repeat-indel alleles 0.1, and 5mC alleles nothing; a read was tagged only if the majority fraction exceeded 0.65 (`-readConfidence`) and the total weighted evidence exceeded 1. The allele counts of the two read populations were then used to re-derive each variant's phase: the fraction of tagged reads consistent with the assignment $\rho = \max(n_1, n_2)/(n_1 + n_2)$, where n_1 and n_2 count haplotype-1 reads carrying the reference allele plus haplotype-2 reads carrying the alternate allele and vice versa, had to exceed 0.75 (`-snpConfidence`); otherwise the variant was left unphased. This second pass removes assignments driven by a small number of chimeric or mis-mapped reads.

Phasing entropy and graph export

For each phased variant we recorded the two weighted haplotype totals h_1 and h_2 (after the single-read guard, where it fired) and their normalized Shannon entropy

$$\text{PE}(v) = - \sum_{i \in \{1,2\}} p_i \log_2 p_i, \quad p_i = \frac{h_i}{h_1 + h_2}, \quad (4)$$

which ranges from 0 (unanimous votes) to 1 bit (equal support for both haplotypes), with the convention $0 \log_2 0 = 0$. PE is set to 0 when $h_1 + h_2 = 0$, so a value of exactly 0 denotes either unanimous support or no votes at all; the two cases are distinguished by H1+H2. [The Results section interprets $\text{PE} = 0$ as “unanimous but sparse votes”. State explicitly there, or add a

supplementary panel, what fraction of the $PE = 0$ variants had $H1 + H2 = 0$, since those carry no vote evidence at all rather than unanimous evidence.] PE, h_1 and h_2 are written to the output VCF as the INFO fields PE, H1 and H2. With `-dot`, the per-chromosome phasing graph is written in DOT format as `<prefix>.<chrom>.dot`, one directed edge $u^a \rightarrow v^b$ per accepted voting pair with the haplotype weight accumulated at u at the time the vote was cast as label; these files are the input of the GNN module.

Graph neural network for phase-error detection

The GNN operates on local windows of the phasing graph and predicts, for each variant, the probability that its phase assignment is wrong (Supplementary Figs. 6–8). Only records that are heterozygous and phased in the input VCFs enter the module. Variant records from the SNV/indel, SV and 5mC VCFs were merged and sorted by position so that windows span all co-phased variant classes, and a window was opened around every such variant with $PE \geq 0.8$ (`-pe-threshold`), extending 20 variants (`-window`) on each side of the trigger in that merged list. A window therefore contains at most $2 \times 20 + 1 = 41$ variants and 82 allele nodes. Windows with more than 256 nodes are skipped; at the default window size this cap is never reached and it binds only for `-window` greater than 127. [Either state the cap as inactive at default settings, as done here, or remove it from Supplementary Table 3, where it is currently listed as an operative constraint (“ ≤ 256 nodes”).]

Node features. Each allele node carried 31 features in four groups (Supplementary Tables 1 and 2; architecture in Supplementary Table 3 and Supplementary Fig. 7). *Phasing evidence:* PE; the allele’s genotype index; the reference and alternate genotype indices; whether the variant is the window trigger; whether it shares the trigger’s phase set; the allele index; the signed position offset from the trigger normalized by the largest absolute offset in the window; h_1 , h_2 , $h_1 + h_2$, the vote ratio $\max(h_1, h_2)/(h_1 + h_2)$; $\log_2(1 + h_1 + h_2)$, $\ln(1 + h_1)$, $\ln(1 + h_2)$; and $\log_2$ of the total vote weight divided by its mean over the 21 variants centred on the variant. Here h_1 and h_2 were taken from the first outgoing DOT edge of each allele when one existed and from INFO/H1, INFO/H2 otherwise. *Genomic context* (computed from the reference when `-r` is given, and zero otherwise): homopolymer length at the variant (capped at 20); GC content in ± 50 bp; the longest tandem-repeat unit count with unit size 2–4 in ± 15 bp, divided by 10 and capped at 1; Shannon entropy of the ± 15 bp sequence; the number of variants within ± 500 bp, minus one and divided by 20; and $\log_{10}(1 + d)$ for the distance d to the nearest indel, capped at $\log_{10}(10001)$. *Graph structure:* the number of distinct target positions of the variant’s outgoing DOT edges, divided by 10 and capped at 1; $\ln(1 + n_{PS})/\ln(101)$ for the number n_{PS} of variants in the variant’s phase set; $\log_{10}(1 + |\text{offset}|)$ from the trigger; the standard deviation of the variant’s outgoing DOT edge weights, divided by 100 and capped at 1; and an indicator of whether the variant is a bridge vertex, i.e. whether deleting it increases the number of connected components of the position-level, undirected window graph. *Variant class:* four one-hot indicators (SNV, indel, SV, 5mC), the SNV/indel distinction being made on $||\text{ALT}| - |\text{REF}|| > 0$.

[Two feature-table errors to correct in Supplementary Table 1, both verified against `GNNProcess.cpp`: 427–451 (feature vector `f[0]–f[30]`). (i) Features 14 and 15 (`log_h1`, `log_h2`) are natural logarithms, `std::log1p`, not `log2` as the table and earlier drafts stated; only feature 13 (`log_total`)

uses `std::log2`. The text above now says `ln`. (ii) The indices of four features are permuted in the table: the code order is 21 `out_degree`, 22 `ps_block_size`, 23 `log_center_dist`, 24 `dist_near_inde1` (1-based), whereas the table lists them as 22, 23, 24 and 21. Re-number the table; the grouping and the definitions are otherwise correct.]

Edge features. Edges were taken from the DOT graph, restricted to pairs both of whose end-points lie inside the window, symmetrized by copying the features of whichever direction was present, and augmented with a self-loop per node whose features are the mean of its non-self incident edges. Each edge carried six features: the logistic-transformed vote weight $1/(1 + e^{-w})$; $\log_{10}(1 + \text{genomic distance})$; $\ln(1 + w)$; whether the two endpoints carry the same allele index; whether the two variants belong to different phase sets; and w divided by the total outgoing weight of the source allele node within the window. [Earlier drafts glossed the fourth feature as “whether the two alleles are on the same haplotype”. It is an allele-index match, not a haplotype match (`GNNProcess.cpp:481, e.src_allele == e.dst_allele`); corrected above, and Supplementary Table 2 already states it correctly.] [Stale code comment, harmless but confusing to reimplementers: the header of `GNNModel.h` (lines 6–7 and 43) describes the edge tensor as `[N, N, 7]` and the per-layer edge encoder as `[7, 128]`, whereas `kEdgeFeat` is 6 (`GNNWeights.h:21`) and six features are what the code writes. Fix the comment.]

Architecture. The network follows the general, powerful, scalable (GPS) graph-transformer recipe⁶⁰. Node features were standardized with a batch-normalization layer⁷⁷ whose statistics are frozen at inference, linearly projected to a 128-dimensional embedding and passed through a GELU non-linearity⁷⁸. Four identical layers then each combine a local and a global branch. The local branch is a four-head GATv2 message-passing layer^{57,58} in which the attention logit of head t for edge $j \rightarrow i$ is

$$\alpha_{ij}^{(t)} \propto \exp\left(\mathbf{a}_t^\top \text{LeakyReLU}_{0.2}([\mathbf{W}_r \mathbf{x}_i + \mathbf{W}_l \mathbf{x}_j + \mathbf{W}_e \mathbf{e}_{ij}]_t)\right), \quad (5)$$

where $[\cdot]_t$ selects the 32 coordinates of head t , $\mathbf{W}_e$ is a per-layer edge encoder without bias and the normalization is a softmax over the incoming edges of i (a node with no incoming edge instead receives the uniform average of $\mathbf{W}_l \mathbf{x}_j$ over all window nodes); the head outputs are concatenated and a bias is added. The global branch is four-head scaled dot-product self-attention over all nodes of the window⁵⁹, with no masking. The two branch outputs are added to the residual stream, followed by layer normalization⁷⁹, a two-layer feed-forward block (128→256→128, GELU) with a second residual connection and layer normalization. The final node embedding is concatenated with two edge-context scalars, the sum and the non-negative maximum of the raw weights of the node’s incoming edges excluding its self-loop, and passed through a two-layer classifier (130→128→2) with softmax output.

The model comprises 687,358 float32 values (2,749,432 bytes), namely 124 in the input batch-normalization layer (scale, shift and the two frozen running statistics over 31 features), 4,096 in the input projection, 166,528 in each of the four GPS layers, and 17,026 in the classifier. Inference is implemented in plain C++ with dense $N \times N$ adjacency and edge tensors and no external runtime dependencies; the exact erf-based GELU must be used, because the tanh approximation changes the output by $\sim 10^{-3}$ and does not reproduce the reference implementation. Predictions for the two allele nodes of a variant were averaged. Only variants in

the central zone of each window were retained, that is those whose normalized absolute offset r from the trigger satisfies $r \leq \min(1, 10/\max(1, \text{round}(20r_{\max})))$, where $r_{\max}$ is the largest normalized absolute offset in the window; because offsets are normalized by that same largest offset, $r_{\max} = 1$ for every window containing a variant more than 1 bp from the trigger, so in practice the retained zone is the central half of the window’s genomic span. A variant covered by several windows received the running mean of its predictions. [Two points on the central-zone rule, verified at `GNNProcess.cpp:517–531`. (i) The code uses `std::round`, not the ceiling that earlier drafts wrote; corrected above. (ii) Because $r_{\max} \equiv 1$ in essentially every window, the expression collapses to the constant 0.5 and the $r_{\max}$ dependence is dead. Confirm that this matches `_center_mask` in the training-time `prepare_gnn_data.py`, and if the intent was a variant-index rather than a genomic-distance criterion, say so; otherwise simplify both the code and Supplementary Fig. 6c to “the central half of the window span”.]

Training data and labels. [Required, and currently absent. State: (i) the sample(s) and the exact alignment files used to generate training windows, and whether they are the same HG002 down-samplings used for evaluation; (ii) the chromosome-level split into training, validation and test sets, given by name, together with the argument that no window straddles a split boundary and that no evaluation chromosome contributed any window, any feature normalization statistic or any hyperparameter choice — without an explicit chromosome hold-out the evaluation in Figs. 2–6 is circular, and a reviewer will say so; (iii) the coverages at which training windows were generated and whether the down-sampling seeds used for training are disjoint from the ten evaluation seeds at 10–20×; (iv) the variant-class configurations from which windows were drawn (SNV-only, SNV+indel, four-class), since the four class-indicator features are only informative if more than one configuration was seen; (v) the number of windows and the number of labelled allele nodes in each split.]

[Label definition, required. State the truth set used to derive labels (GIAB v4.2.1 or the Q100/v5.0q benchmark) and the exact rule: presumably a variant is labelled positive when, after orienting each phase block to the truth haplotypes by the majority assignment, its assignment disagrees with the truth, i.e. it participates in a switch or a flip. Specify the treatment of (a) variants absent from the truth set, (b) variants homozygous in the truth set, and (c) indels, SVs and 5mC sites, for which no switch-error truth exists — Fig. 6 shows that most unphased variants are benchmark-absent, so whether such variants were labelled, discarded or labelled positive materially changes what the network learned.]

[Class balance, required. Report the positive-class prevalence among labelled nodes and state any resampling, class weighting or focal weighting applied. At the switch-error rates reported in Fig. 2b the raw prevalence is well below 1%, so an unweighted cross-entropy loss on the raw distribution would be a surprising choice and must be justified or corrected.]

Training procedure. [Required, one value each: loss function (presumably cross-entropy) and any class weights; optimizer (presumably AdamW⁸⁰) with β_1 , β_2 , ε and weight decay; peak learning rate and schedule (warm-up, decay); batch size in windows or in nodes, and whether windows were batched as disjoint graphs; number of epochs trained; gradient clipping; dropout, if any (none is present in the inference graph, so state that it was either absent or disabled at export); random seed(s) and whether results are robust to reseeding; deep-learning framework

and version, CUDA/cuDNN versions; GPU or CPU model and count; total wall-clock training time; and the wall-clock time and peak memory of the feature-extraction step that produced the training windows.]

Model selection and export. The released weights are the checkpoint `model_e6.onnx`, recorded in the header of the generated weight file (`GNNWeights.h`, lines 3–4 at commit `cc17fb1`). Training was performed in Python, the checkpoint was exported to ONNX, and the ONNX initializers were converted by `embed_weights.py` into a base64-encoded little-endian float32 blob compiled into the binary as `GNNWeights.h`; the C++ forward pass in `GNNModel.h` reads the blob in a fixed tensor order and refuses to run if its length disagrees with `kParamCount`. [Required: the validation criterion and the epoch-6 selection rule (best validation loss, best validation AUPRC, best downstream switch-error reduction, or a fixed budget); the validation metric value at the selected checkpoint and at the neighbouring checkpoints; and a statement of the numerical agreement achieved between the PyTorch model, the exported ONNX graph and the C++ implementation on a fixed set of windows (e.g. maximum absolute difference in the predicted error probability), since the correctness of every reported `gnn` result depends on that agreement.] [Neither `prepare_gnn_data.py` nor `embed_weights.py` nor any training script, configuration file, checkpoint or ONNX model is present anywhere in the repository at `cc17fb1` or on any other branch; the C++ sources reference them only in comments (`GNNProcess.cpp`:355–357, `GNNProcess.h`:108, `GNNWeights.h`:1). They must be deposited before submission (see Code availability).]

Overfitting and generalization. [State explicitly, in one sentence each: what was held out (chromosomes, coverages, seeds, variant-class configurations) and what was not. Then state plainly what has not been tested: the network was trained and all results were obtained on a single individual (HG002) sequenced on a single platform and chemistry, so no independent-sample and no independent-platform evaluation exists. Two mitigating design properties may be stated as design choices rather than as evidence: the features contain no sequence embedding and no sample-specific statistic, and the only permitted action is to withhold phase, so a miscalibrated model costs contiguity rather than creating haplotypes the graph did not support. A reviewer will nonetheless ask for at least one of: HG001 or HG005 nanopore data, PacBio HiFi data, or a second nanopore chemistry, scored with the identical pipeline and the identical embedded weights.]

Correction. A variant whose mean predicted error probability was ≥ 0.30 (`-break-threshold`) was unphased: its genotype was rewritten in unordered, ascending form (`0|1` and `1|0` both become `0/1`) and its PS tag was removed, while the pair of alleles was preserved and no genotype was ever flipped. Bridge vertices may optionally be protected (`-respect-bridge`; off by default). After unphasing, the remaining phased variants of each phase set were checked for connectivity through the surviving DOT edges; when a phase set became disconnected, the largest component kept the original PS and every other component received a new PS equal to the 1-based position of its first variant (`-no-split-blocks` disables this). The corrected SNV/indel, SV and 5mC VCFs are written separately, as `<prefix>.vcf`, `<prefix>_SV.vcf` and `<prefix>_mod.vcf`.

Haplotagging

Reads were assigned to haplotypes by **haplotag**, which counts, for each primary (optionally supplementary, **-tagSupplementary**) alignment with mapping quality ≥ 1 (**-qualityThreshold**), the phased SNV, indel, SV and 5mC alleles it carries on each haplotype and tags the read with **HP:i:1** or **HP:i:2** and the phase set **PS** when the majority haplotype accounts for at least 60% of the informative alleles (**-percentageThreshold**); reads without a clear majority are left untagged. SV alleles were re-derived from the CIGAR string with the same window and length tolerance as in **phase**. Tagging can be restricted to a region (**-region**) and a per-read log can be written (**-log**). Output can be written as BAM or CRAM (**-cram**).

Benchmark data

[Confirm and complete; every field below marked **todo** is mandatory for a Nature Methods Methods section and none may be guessed.] HG002 (NA24385) Oxford Nanopore R10.4.1 reads ([flow cell and kit; sequencing centre; basecaller name, version and model, e.g. Dorado v... sup@v... with the 5mCG_5hmCG modification model; whether simplex or duplex; N50 read length; source URL and, if one exists, the SRA/ENA/GIAB accession — a URL alone is not an accession]) were aligned to GRCh38 ([exact analysis set, e.g. GCA_000001405.15_GRCh38_no_alt_analysis_set.fna with or without decoys, and the accession]) with minimap2 ([version]) using **-ax map-ont -y**. The full data set ([mean autosomal coverage, computed how]) was down-sampled with **samtools view -s**⁸¹ ([samtools version]) to 10, 12, 14, 16, 18, 20, 30, 40, 50 and 60×; for 10–20×, ten independent random seeds were drawn and all metrics are reported as the mean over the ten replicates, whereas 30–60× used a single replicate ([give the ten seed values, or state that they were 1–10, so the down-sampling is reproducible]). SNVs and indels were called with Clair3 ([version and model directory, e.g. r1041_e82_400bps_sup_v...; and whether calls were made once on the full-coverage BAM or separately at each down-sampled coverage — these give different results and the choice must be stated]), SVs with Sniffles2 ([version]) with **-output-rnames**, and allele-specific 5mC sites with longphase modcall ([state whether modcall was run per coverage and with which options]). Phasing accuracy was assessed against two truth sets: the GIAB v4.2.1 small-variant benchmark for HG002⁵³ ([benchmark VCF and BED filenames and the GIAB FTP path]), and the T2T-based HG002 Q100 benchmark^{54–56} ([exact release, e.g. v1.1; the manuscript calls it “v5.0q” throughout — either define that name here on first use or replace it with the official release identifier, because “v5.0q” is not a name a reader can look up]), which extends the evaluated regions into segmental duplications, satellite repeats and other regions excluded from v4.2.1. Identical phased VCFs were scored against both truth sets. [State whether evaluation was restricted to the benchmark high-confidence BED of each truth set, and whether chrX, chrY and chrM were included; the phased-fraction denominators in Figs. 2a and 4a depend on both.]

Comparison with other methods

WhatsHap v2.8¹³ and HapCUT2 v1.3.4¹⁴ were run on the same alignments and the same Clair3 calls ([give the complete command line for each, including WhatsHap **phase** options such as

`-ignore-read-groups` and any `-indels/-distrust-genotypes` setting, and the HapCUT2 two-step invocation `extractHAIRS -ont 1` followed by HAPCUT2; “default parameters” is not reproducible for either tool]). LongPhase 1.0¹⁶ ([exact version and release tag; Supplementary Table 7 currently says “1.0.x”]) and LongPhase 2 were run with `-ont`; the GNN module was applied to the LongPhase 2 output with default thresholds ($PE \geq 0.8$, break threshold 0.30, no bridge protection, block splitting enabled). For co-phasing experiments, the SNV, SNV+indel, SNV+SV, SNV+5mC and SNV+indel+SV+5mC configurations were phased separately, and the GNN was applied to each. Because WhatsHap and HapCUT2 phase only small variants, cross-tool comparisons were restricted to SNVs; indel, SV and 5mC co-phasing were evaluated as LongPhase 2 with and without GNN correction. [MethPhaser⁵⁰ is named here but no MethPhaser result appears in any figure or in the Results text. Either add the methylation-assisted comparison (version, command line, and which SNV phaser supplied its input) or delete this sentence; as it stands the Methods promises a comparison the paper does not report, which reviewers check.] [State that all tools were given identical inputs and run on the same machine, and whether any tool was re-run per down-sampling replicate.]

Evaluation metrics

Metrics (Supplementary Fig. 10) were computed with `longphase compare`, a C++ re-implementation of the `whatshap compare` algorithm¹³ that is parallelized over chromosomes with OpenMP. For each chromosome, heterozygous variants common to the truth and query VCFs were intersected by position and alleles and partitioned into blocks that are phased in both files; singleton blocks were ignored. Within each block the query haplotype string was compared with the truth string in switch encoding (the sequence of relative phase changes between consecutive variants), giving the number of *switch errors*; the *switch error rate* is the number of switch errors divided by the number of assessed variant pairs. The *Hamming distance* is the block-wise minimum over the two possible haplotype orientations of the number of misassigned variants, summed over blocks and expressed as a percentage of compared variants. Switch errors were further decomposed into isolated switches and flips (two consecutive switches). The *phased fraction* is the number of truth heterozygous variants phased by the query divided by the total number of truth heterozygous variants, reported separately for SNVs, indels, SVs and 5mC sites; the *block N50* is the span-weighted median block length of the query over all chromosomes. Switch-error rates and Hamming distances for cross-tool comparisons were computed on SNVs only, so that the co-phasing of additional variant classes did not change the evaluated set. Switch-error positions were exported as BED for the analysis of unphased variants, in which each variant unphased by the GNN was classified as absent from the benchmark, homozygous in the benchmark, or a true heterozygous site, and cross-tabulated by read depth and phasing informativeness. [Required for the claim that `longphase compare` “yields identical values” to `whatshap compare`: report the concordance explicitly (e.g. identical switch-error and Hamming counts on all autosomes for all four tools at one coverage, with the `whatshap compare` version used), in a supplementary table. As previously written the equivalence was asserted without evidence.] [Define the depth and informativeness bins used in Fig. 6b (“low”, “normal”, “high” relative to the genome-wide mean, and the informativeness cut-points) and the “low genotype quality” threshold used in

Fig. 6c.]

Runtime and memory

Wall-clock time and peak resident memory were measured with `/usr/bin/time -v` on [CPU model and clock, physical core and socket count, RAM, storage type, operating system and kernel, compiler and optimization flags], using $[t]$ threads for every tool. [State whether the machine was otherwise idle, how many repeats were timed and whether the reported value is the mean, median or minimum, and report peak resident memory alongside every runtime — the Results section currently quotes runtimes only, and Fig. 2e has no memory counterpart.] GNN correction time was measured separately and added to the LongPhase 2 total. Reported LongPhase 2 times include reading of the alignments, graph construction, phasing and writing of the VCF, and exclude variant calling, alignment and down-sampling.

Statistics and reproducibility

No statistical tests were used to compare methods, and no sample-size calculation, randomization or blinding was applicable: all metrics are deterministic functions of the input data and the only stochastic element is the choice of reads during down-sampling. The replicate design was as follows. At 10, 12, 14, 16, 18 and 20 $\times$, ten independent down-sampling replicates were generated from the same full-coverage alignment with ten distinct random seeds, and every reported value at those coverages is the mean over the ten replicates. At 30, 40, 50 and 60 $\times$, a single replicate was generated and analysed, so no dispersion can be reported at those coverages and differences between methods there are not accompanied by a measure of replicate variability. [Required before submission: (i) report dispersion for every 10–20 $\times$ value quoted in the Results and show it in Figs. 2–5, as either s.d. or the full range over the ten replicates, and say in each figure legend which of the two is drawn and that $n = 10$ down-sampling replicates; (ii) either generate at least two further replicates at 30 $\times$ and 60 $\times$ so that the high-coverage ranking carries a dispersion estimate, or state explicitly in the Results and in the legends of Figs. 2–5 that the 30–60 $\times$ points are single replicates and that the method ranking there is unreplicated.] Phasing results were verified to be identical across thread counts [state the thread counts compared and how identity was established, e.g. identical MD5 of the output VCFs at $t = 32$ and $t = 64$], and the GNN module is deterministic given its input VCF and DOT files because the trained weights are fixed and inference performs no sampling. [Confirm that the per-variant running mean over overlapping windows in `GNNProcess.cpp`:531–537 is thread-order independent; it is a floating-point running mean accumulated under a mutex in completion order, so its last bits may depend on thread scheduling. If it does vary, either state the magnitude as negligible relative to the 0.30 break threshold or make the accumulation order-independent.]

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article. [The Reporting Summary must be completed and submitted; its statistics section must match the replicate design stated above ($n = 10$ down-sampling replicates at

10–20 $\times$, $n = 1$ at 30–60 $\times$), and its “Data” section must list every accession given below.]

Data availability

[Required, with the exact identifier in each case. (i) HG002 ONT R10.4.1 reads: repository and accession (SRA/ENA study and run accessions, or the GIAB FTP path and file names with their release date); state the licence or data-use terms. (ii) GRCh38 reference: exact analysis-set file and its accession. (iii) GIAB HG002 v4.2.1 small-variant benchmark: FTP URL, VCF and BED file names. (iv) HG002 Q100/T2T benchmark: release identifier, DOI or FTP URL and file names. (v) Sniffles2 SV calls, Clair3 small-variant calls, `modcall` 5mC calls, all phased VCFs for all tools, coverages and replicates, the DOT graph exports, the GNN training/validation/test window sets, and the `longphase compare` output tables underlying every figure panel: deposit in Zenodo and give the concept DOI (which resolves to all versions) and the version DOI, together with the licence (CC BY 4.0 is the usual choice). (vi) Source data for Figs. 2–6 and for every Supplementary Figure with plotted values. Nature Methods requires that restrictions, if any, are stated; if there are none, say so explicitly.]

Code availability

LongPhase 2, including the `modcall`, `phase`, `gnn`, `haplotag` and `compare` sub-commands, is open source under the GNU General Public License v3.0 at <https://github.com/twolinin/longphase>. A Dockerfile, pre-compiled Linux x86-64 binaries and a demonstration data set (chr1:121.5–122.0 Mb, `demo/demo.sh`, which runs `modcall`, SNV-only and four-class `phase`, `gnn` correction of both and `compare` against a truth VCF in under two minutes on four threads) are provided in the repository. [Give the exact release tag and the full 40-character commit hash of the version used for every result in this paper, and archive that release in Zenodo with its own DOI. The latest tagged release at the time of writing is v2.0.2 (`main.cpp`:9; `README.md`:34), but v2.0.2 predates the `gnn` and `compare` sub-commands, which exist only on the unmerged JH branch at commit `cc17fb1` — so no released version can currently reproduce any `gnn` or `compare` result in this paper. Merge and tag before submission.] [The training code is not public and is not in the repository: `prepare_gnn_data.py` (window and feature extraction), the training script itself, the ONNX export, `embed_weights.py` (which generates `GNNWeights.h`), the `model_e6` checkpoint and the environment specification (framework, CUDA and package versions) must all be deposited, under an OSI-approved licence, in the repository or in the Zenodo archive, and cited here by DOI. Nature Methods will not accept a machine-learning method whose training code is unavailable.] [Deposit the analysis scripts that produced Figs. 2–6 from the `longphase compare` tables, including `latex/figures/supp/make_supp_figs.py`.]

References

- [1] Tewhey, R., Bansal, V., Torkamani, A., Topol, E. J. & Schork, N. J. The importance of phase information for human genomics. *Nature Reviews Genetics* **12**, 215–223 (2011).
- [2] Snyder, M. W., Adey, A., Kitzman, J. O. & Shendure, J. Haplotype-resolved genome

sequencing: experimental methods and applications. *Nature Reviews Genetics* **16**, 344–358 (2015).

[3] Sun, Q. & Li, Y. Advances in haplotype phasing and genotype imputation. *Nature Reviews Genetics* **27**, 155–169 (2025).

[4] Li, Q. *et al.* Unraveling the hidden complexity of cancer through long-read sequencing. *Genome Research* **35**, 599–620 (2025).

[5] Logsdon, G. A., Vollger, M. R. & Eichler, E. E. Long-read human genome sequencing and its applications. *Nature Reviews Genetics* **21**, 597–614 (2020).

[6] Wenger, A. M. *et al.* Accurate circular consensus long-read sequencing improves variant detection and assembly of a human genome. *Nature Biotechnology* **37**, 1155–1162 (2019).

[7] Beyter, D. *et al.* Long-read sequencing of 3,622 Icelanders provides insight into the role of structural variants in human diseases and other traits. *Nature Genetics* **53**, 779–786 (2021).

[8] Kolmogorov, M. *et al.* Scalable Nanopore sequencing of human genomes provides a comprehensive view of haplotype-resolved variation and methylation. *Nature Methods* **20**, 1483–1492 (2023).

[9] Gustafson, J. A. *et al.* High-coverage nanopore sequencing of samples from the 1000 Genomes Project to build a comprehensive catalog of human genetic variation. *Genome Research* **34**, 2061–2073 (2024).

[10] and Porubsky, D. *et al.* Fully phased human genome assembly without parental data using single-cell strand sequencing and long reads. *Nature Biotechnology* **39**, 302–308 (2020).

[11] Garg, S. Computational methods for chromosome-scale haplotype reconstruction. *Genome Biology* **22** (2021).

[12] Patterson, M. *et al.* W <scp>hats</scp> h <scp>ap</scp> : Weighted haplotype assembly for Future-Generation sequencing reads. *Journal of Computational Biology* **22**, 498–509 (2015).

[13] Martin, M. *et al.* WhatsHap: fast and accurate read-based phasing. *bioRxiv* (2016).

[14] Edge, P., Bafna, V. & Bansal, V. HapCUT2: robust and accurate haplotype assembly for diverse sequencing technologies. *Genome Research* **27**, 801–812 (2016).

[15] Shafin, K. *et al.* Haplotype-aware variant calling with PEPPER-Margin-DeepVariant enables high accuracy in nanopore long-reads. *Nature Methods* **18**, 1322–1332 (2021).

[16] Lin, J.-H., Chen, L.-C., Yu, S.-C. & Huang, Y.-T. LongPhase: an ultra-fast chromosome-scale phasing algorithm for small and large variants. *Bioinformatics* **38**, 1816–1822 (2022).

[17] Sedlazeck, F. J. *et al.* Accurate detection of complex structural variations using single-molecule sequencing. *Nature Methods* **15**, 461–468 (2018).

- [18] Smolka, M. *et al.* Detection of mosaic and population-level structural variants with Sniffles2. *Nature Biotechnology* **42**, 1571–1580 (2024).
- [19] Jiang, T. *et al.* Long-read-based human genomic structural variation detection with cuteSV. *Genome Biology* **21** (2020).
- [20] Zheng, Z. *et al.* Symphonizing pileup and full-alignment for deep learning-based long-read variant calling. *Nature Computational Science* **2**, 797–803 (2022).
- [21] Zheng, Z. *et al.* Accelerated long-read variant calling with Clair3 for whole-genome sequencing. *Bioinformatics* **42** (2026).
- [22] Zheng, Z. *et al.* Clair3-RNA: a deep learning-based small variant caller for long-read RNA sequencing data. *Nature Communications* **16** (2025).
- [23] Zheng, Z. *et al.* ClairS: a deep-learning method for long-read tumor–normal pair somatic small variant calling. *Nature Methods* **23**, 1507–1520 (2026).
- [24] Chen, L. *et al.* ClairS-TO: a deep-learning method for long-read tumor-only somatic small variant calling. *Nature Communications* **16** (2025).
- [25] Kolesnikov, A. *et al.* Local read haplotagging enables accurate long-read small variant calling. *Nature Communications* **15** (2024).
- [26] Keskus, A. G. *et al.* Severus detects somatic structural variation and complex rearrangements in cancer genomes using long-read sequencing. *Nature Biotechnology* **44**, 247–257 (2025).
- [27] Oxford Nanopore Technologies. `epi2me-labs/wf-human-variation`: Nextflow workflow for the analysis of human genetic variation from Oxford Nanopore sequencing data. <https://github.com/epi2me-labs/wf-human-variation> (2025). Version 2.x; phasing performed with LongPhase (`-use_longphase`).
- [28] Pacific Biosciences. HiFi-somatic-WDL: tumor-normal variant calling workflow using HiFi reads. <https://github.com/PacificBiosciences/HiFi-Somatic-WDL> (2025).
- [29] Gabbutt, C. *et al.* Fluctuating DNA methylation tracks cancer evolution at clinical scale. *Nature* **645**, 764–773 (2025).
- [30] O’Neill, K. *et al.* Long-read sequencing of an advanced cancer cohort resolves rearrangements, unravels haplotypes, and reveals methylation landscapes. *Cell Genomics* **4**, 100674 (2024).
- [31] Diaz, M. *et al.* Novel driver gene MDC1 confers homologous recombination repair deficiency and genomic instability in chemoresistant relapsing ovarian cancer. *Journal of Translational Medicine* **24** (2026).
- [32] McAuley, L. *et al.* Whole genome sequencing of pre-treatment and post-treatment locally advanced rectal cancer using long and short read technologies. *Scientific Reports* **16** (2026).

- [33] Zalusky, M. P. *et al.* 3-hour genome sequencing and targeted analysis to rapidly assess genetic risk. *Genetics in Medicine Open* **2**, 101833 (2024).
- [34] Smits, D. J. *et al.* Nanopore long-read sequencing for the critically ill facilitates ultrarapid diagnostics and urgent clinical decision making. *European Journal of Human Genetics* **34**, 108–118 (2025).
- [35] LaFlamme, C. W. *et al.* Diagnostic utility of DNA methylation analysis in genetically unsolved pediatric epilepsies and CHD2 epismature refinement. *Nature Communications* **15** (2024).
- [36] Iwasaki, Y. *et al.* Bidirectional disruption of *GNAS* transcripts causes broad methylation defects in pseudohypoparathyroidism type 1b. *Proceedings of the National Academy of Sciences* **122** (2025).
- [37] Thomas, R. *et al.* Splice-modulating antisense oligonucleotides targeting a pathogenic intronic variant in adult polyglucosan body disease correct mis-splicing and restore enzyme activity in patient cells. *Nucleic Acids Research* **53** (2025).
- [38] Tian, Y., McDonnell, S. K., Wu, L., Larson, N. B. & Wang, L. Fine mapping regulatory variants by characterizing native CpG methylation with nanopore long-read sequencing. *Human Genetics and Genomics Advances* **7**, 100532 (2026).
- [39] Wang, S. *et al.* A scalable framework for comprehensive typing of polymorphic immune genes from Long-Read data. *Advanced Science* **13** (2026).
- [40] Galvez, J. H. *et al.* Benchmarking of Oxford Nanopore whole genome sequencing for germline variant and CpG methylation detection across Canada’s national platform for genome sequencing and analysis. *Frontiers in Genetics* **17** (2026).
- [41] Flack, N. *et al.* Chromosome-level, nanopore-only genome and allele-specific DNA methylation of Pallas’s cat, *Otocolobus manul*. *NAR Genomics and Bioinformatics* **5** (2023).
- [42] Mastroianni, F. K., Miller, D. E. & Eichler, E. E. Applications of long-read sequencing to Mendelian genetics. *Genome Medicine* **15** (2023).
- [43] Mahmoud, M., Agostinho, D. P. & Sedlazeck, F. J. A Hitchhiker’s guide to long-read genomic analysis. *Genome Research* **35**, 545–558 (2025).
- [44] Zheng, X., Sanio, P., Luo, R., Li, H. & Sedlazeck, F. J. Bioinformatics for human long-read whole-genome sequencing. *Nature Reviews Methods Primers* **6** (2026).
- [45] Holt, J. M. *et al.* HiPhase: jointly phasing small, structural, and tandem repeat variants from HiFi sequencing. *Bioinformatics* **40** (2024).
- [46] Battistella, E., Maheshwari, A., Ekim, B., Berger, B. & Popic, V. Graph-based deep reinforcement learning for haplotype assembly with Ralphi. *Genome Research* **35**, 2617–2625 (2025).

- [47] Luo, J., Wang, J., Zhai, H. & Wang, J. GCphase: an SNP phasing method using a graph partition and error correction algorithm. *BMC Bioinformatics* **25** (2024).
- [48] Cao, S. *et al.* cutehap: Haplotype-Aware structural variant detection in phased Long-Read sequencing data. *Advanced Science* **13** (2026).
- [49] Akbari, V. *et al.* Megabase-scale methylation phasing using nanopore long reads and NanoMethPhase. *Genome Biology* **22** (2021).
- [50] Fu, Y. *et al.* MethPhaser: methylation-based long-read haplotype phasing of human genomes. *Nature Communications* **15** (2024).
- [51] Chen, H., Nie, F., Luo, F. & Wang, J. HapBridge: A Methylation-Guided approach for correcting switch errors and bridging phased blocks in Long-Read phasing. *bioRxiv* (2025).
- [52] Zook, J. M. *et al.* An open resource for accurately benchmarking small variant and reference calls. *Nature Biotechnology* **37**, 561–566 (2019).
- [53] Wagner, J. *et al.* Benchmarking challenging small variants with linked and long reads. *Cell Genomics* **2**, 100128 (2022).
- [54] Jarvis, E. D. *et al.* Semi-automated assembly of high-quality diploid human reference genomes. *Nature* **611**, 519–531 (2022).
- [55] Rhie, A. *et al.* The complete sequence of a human Y chromosome. *Nature* **621**, 344–354 (2023).
- [56] Genome in a Bottle Consortium. HG002 Q100 T2T diploid assembly-based small-variant benchmark (draft v1.x / GIAB v5). https://ftp-trace.ncbi.nlm.nih.gov/ReferenceSamples/giab/release/AshkenazimTrio/HG002_NA24385_son/T2T-Q100/ (2025).
- [57] Veličković, P. *et al.* Graph attention networks. In *International Conference on Learning Representations (ICLR)* (2018).
- [58] Brody, S., Alon, U. & Yahav, E. How attentive are graph attention networks? In *International Conference on Learning Representations (ICLR)* (2022).
- [59] Vaswani, A. *et al.* Attention is all you need. In *Advances in Neural Information Processing Systems*, vol. 30 (2017).
- [60] Rampášek, L. *et al.* Recipe for a general, powerful, scalable graph transformer. In *Advances in Neural Information Processing Systems*, vol. 35 (2022).
- [61] Ho, M.-E., Zheng, Z., Luo, R., Chiang, H.-H. & Huang, Y.-T. LongPhase-s: purity estimation and variant recalibration with somatic haplotyping for long-read sequencing. *bioRxiv* (2025).
- [62] Chen, Z.-Y. *et al.* LongPhase-to: somatic haplotype reconstruction and variant recalibration from tumor-only long-read sequencing. *Manuscript in review* (2026). Manuscript in review.

- [63] Nurk, S. *et al.* The complete sequence of a human genome. *Science* **376**, 44–53 (2022).
- [64] Liao, W.-W. *et al.* A draft human pangenome reference. *Nature* **617**, 312–324 (2023).
- [65] Vrček, L. *et al.* Geometric deep learning framework for de novo genome assembly. *Genome Research* **35**, 839–849 (2024).
- [66] Park, J. *et al.* DeepSomatic: Accurate somatic small variant discovery for multiple sequencing technologies. *bioRxiv* (2024).
- [67] Ahmad, T. *et al.* Wakhan: reconstruction of chromosome-scale copy number profiles of tumor genomes with long-read sequencing. *medRxiv* (2025).
- [68] Cheng, H., Concepcion, G. T., Feng, X., Zhang, H. & Li, H. Haplotype-resolved de novo assembly using phased assembly graphs with hifiasm. *Nature Methods* **18**, 170–175 (2021).
- [69] Rautiainen, M. *et al.* Telomere-to-telomere assembly of diploid chromosomes with Verkko. *Nature Biotechnology* **41**, 1474–1482 (2023).
- [70] Selvaraj, S., R Dixon, J., Bansal, V. & Ren, B. Whole-genome haplotype reconstruction using proximity-ligation and shotgun sequencing. *Nature Biotechnology* **31**, 1111–1118 (2013).
- [71] Henglin, M. *et al.* Graphasing: phasing diploid genome assembly graphs with single-cell strand sequencing. *Genome Biology* **25** (2024).
- [72] Ebler, J. *et al.* Pangenome-based genome inference allows efficient and accurate genotyping across a wide spectrum of variant classes. *Nature Genetics* **54**, 518–525 (2022).
- [73] Fan, S. *et al.* A long-read based haplotype panel enhances imputation and discovery of functional small and structural variants. *Research Square* (2025).
- [74] Bonfield, J. K. *et al.* HTSlib: C library for reading/writing high-throughput sequencing data. *GigaScience* **10** (2021).
- [75] Oxford Nanopore Technologies. Dorado: high-performance, easy-to-use, open source base-caller for Oxford Nanopore reads. <https://github.com/nanoporetech/dorado> (2025).
- [76] Li, H. Minimap2: pairwise alignment for nucleotide sequences. *Bioinformatics* **34**, 3094–3100 (2018).
- [77] Ioffe, S. & Szegedy, C. Batch normalization: accelerating deep network training by reducing internal covariate shift. In *International Conference on Machine Learning (ICML)* (2015).
- [78] Hendrycks, D. & Gimpel, K. Gaussian error linear units (GELUs). *arXiv* (2016). ArXiv:1606.08415.
- [79] Ba, J. L., Kiros, J. R. & Hinton, G. E. Layer normalization. *arXiv* (2016). ArXiv:1607.06450.

- 1082 [80] Loshchilov, I. & Hutter, F. Decoupled weight decay regularization. In *International Con-*
1083 *ference on Learning Representations (ICLR)* (2019).
- 1084 [81] Danecek, P. *et al.* Twelve years of SAMtools and BCFtools. *GigaScience* **10** (2021).

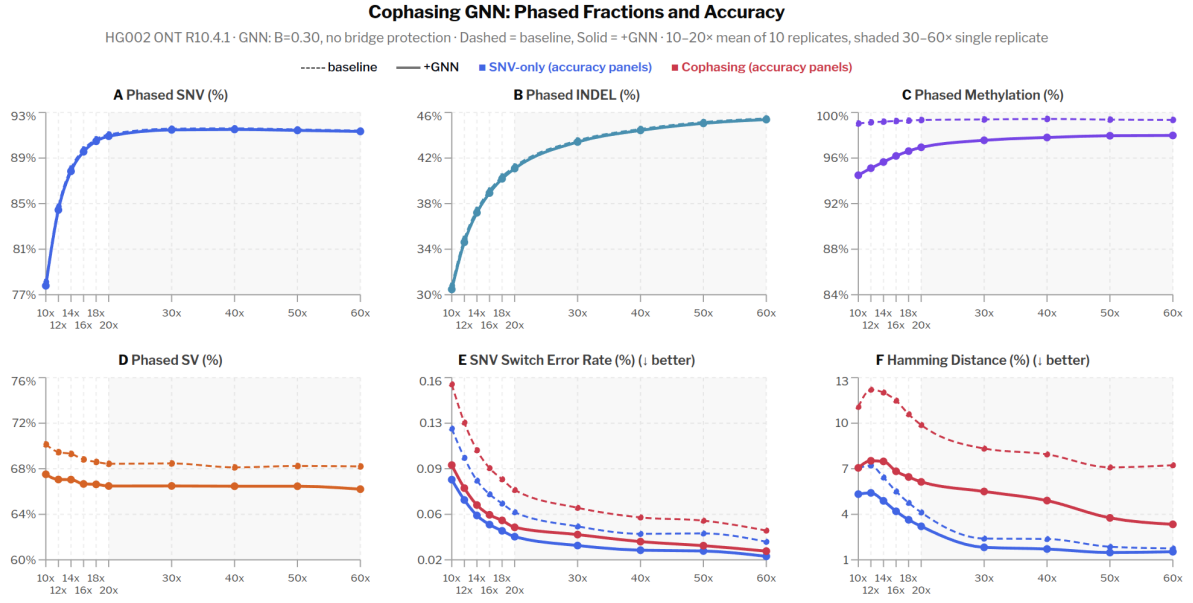


Figure 4: **Co-phasing of SNVs, indels, SVs and allele-specific methylation.** HG002 nanopore R10.4.1 data, 10–60×; dashed lines, LongPhase 2; solid lines, LongPhase 2 with neural-network correction. **a–d**, Fraction of phased SNVs (**a**), indels (**b**), allele-specific 5mC sites (**c**) and SVs (**d**) in the four-class co-phasing configuration. **e,f**, SNV switch error rate (**e**) and Hamming distance (**f**) for SNV-only phasing (blue) and four-class co-phasing (red), each with (solid) and without (dashed) correction. Denominators: GIAB v5.0q heterozygous SNVs and indels; Sniffles2 heterozygous SVs; modcall heterozygous 5mC sites.

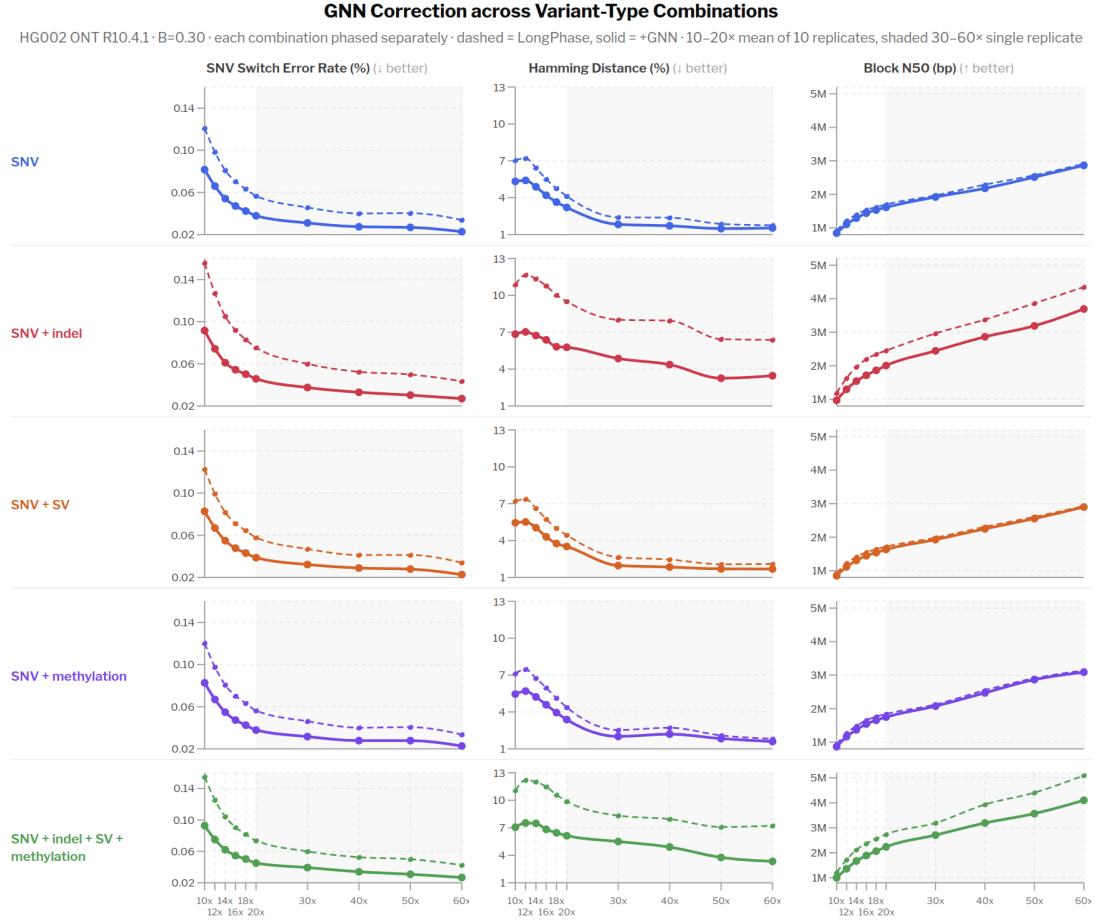
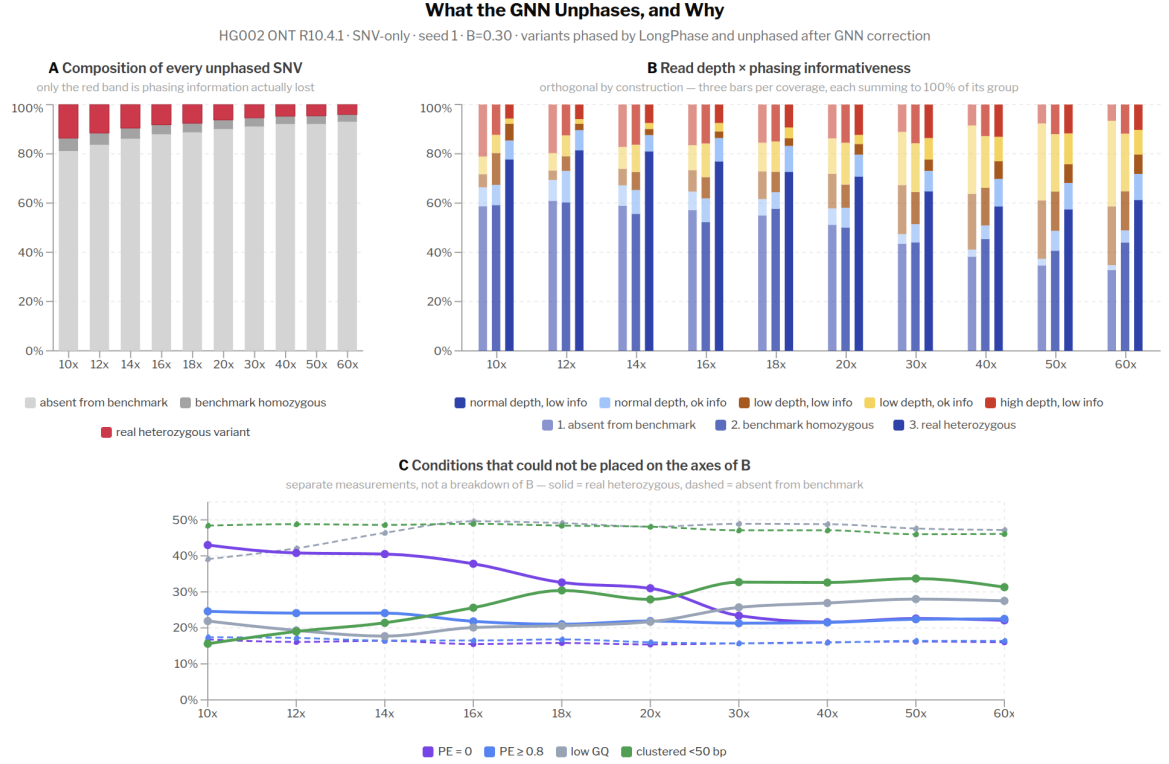


Figure 5: **Effect of neural-network correction across variant-class combinations.** Each row is a separate phasing run of HG002 nanopore R10.4.1 data with the indicated variant classes; dashed, LongPhase 2; solid, LongPhase 2 with correction. Columns: SNV switch error rate, Hamming distance and phase-block N50, scored against the v5.0q benchmark on SNVs. Replicates as in Fig. 2.



The GNN unphases roughly 60,000 SNVs at every coverage, but 86–96% of them are calls the benchmark does not support as heterozygous, so almost none of the apparent loss is real phasing information (A); at 60× only 4.0% (2,411 variants) are genuine heterozygous sites. Placing each variant on two orthogonal axes (B) shows that true heterozygotes are removed overwhelmingly at normal read depth with weak phasing informativeness (61–81%), whereas benchmark-absent calls shift towards low depth as coverage rises (12.5% at 10× to 58.6% at 60×), the signature of hard-to-map regions. No variant is unphased for high depth alone: the high-depth / normal-informativeness cell is empty at every coverage. Panel C reports conditions that overlap too heavily to sit on those axes, as plain prevalences within each group. Among the true heterozygotes the GNN removes, 43.0% at 10× have a PE of exactly zero — against 16.8% of benchmark-absent calls — so no fixed entropy threshold could have found them; that share falls to 22.1% by 60× as phasing evidence improves. Clustering and low genotype quality run the other way, staying near 46–49% among benchmark-absent calls at every coverage while remaining well below that among true heterozygotes, which is what marks them as caller artefacts rather than phasing failures.

Figure 6: Composition of the variants unphased by the neural network. SNV-only phasing of HG002 nanopore R10.4.1 data (seed 1; break threshold 0.30); every SNV phased by LongPhase 2 and unphased after correction is classified against the v5.0q benchmark. **a**, Fraction of unphased SNVs that are absent from the benchmark, homozygous in the benchmark or genuine heterozygous sites; only the last category represents lost phasing information. **b**, Within each of the three categories, distribution over read depth (low, normal, high relative to the genome-wide mean) and phasing informativeness (number of informative linking reads); the three bars at each coverage each sum to 100% of their category. **c**, Prevalence of conditions that overlap the axes of **b**: phasing entropy of exactly 0, phasing entropy ≥ 0.8 , low genotype quality and clustering within 50 bp of another variant; solid, genuine heterozygous sites; dashed, benchmark-absent calls.